# Methods in Human Geography

## Quantitative Methods: Spatial Analysis I




Justin van Dijk
j.t.vandijk@ucl.ac.uk

# This week

Part I

- Spatial data.
- GIScience and GISystems.

Part II

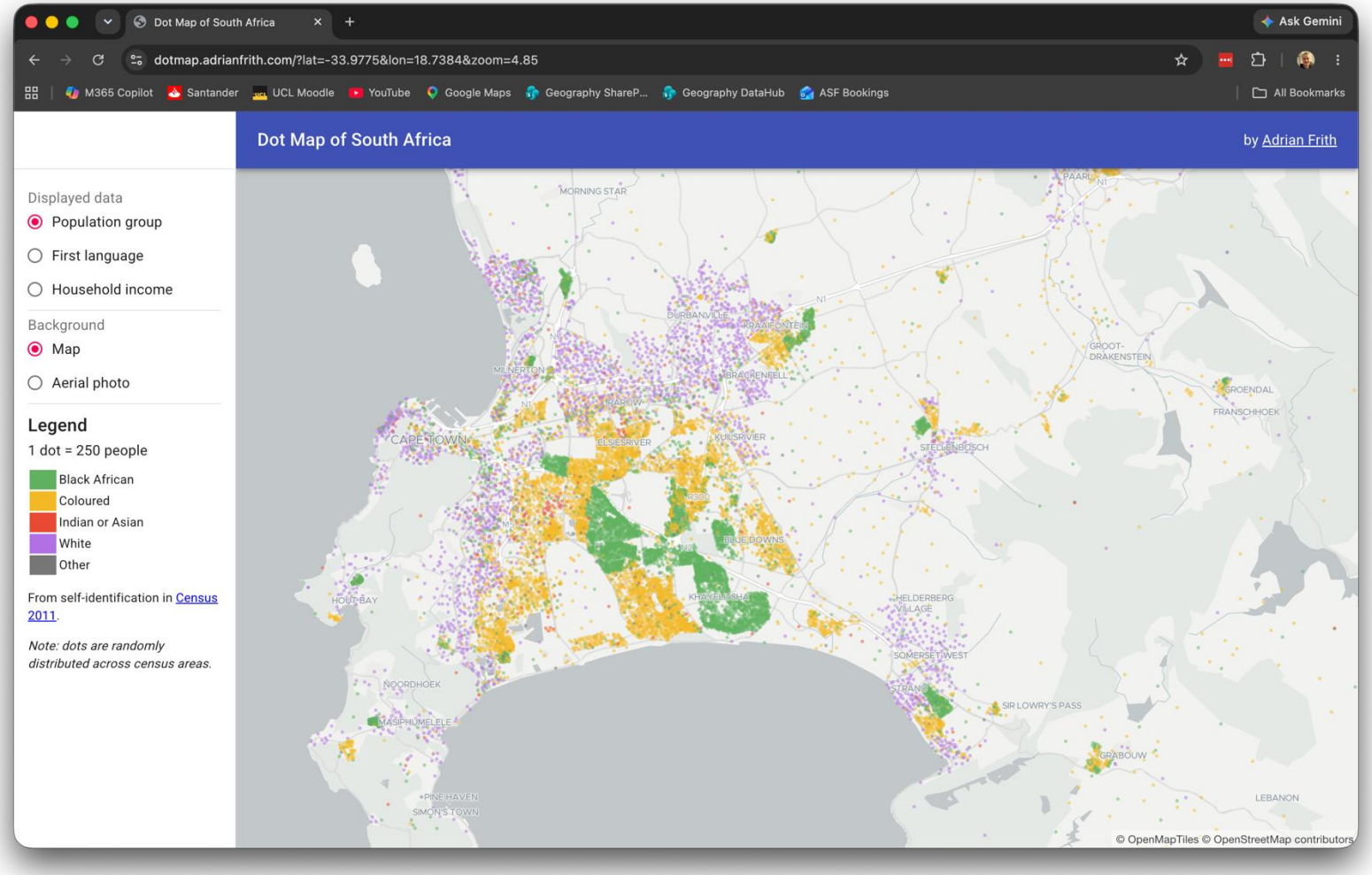
- Data sources.
- Attribute joins.

# Spatial data

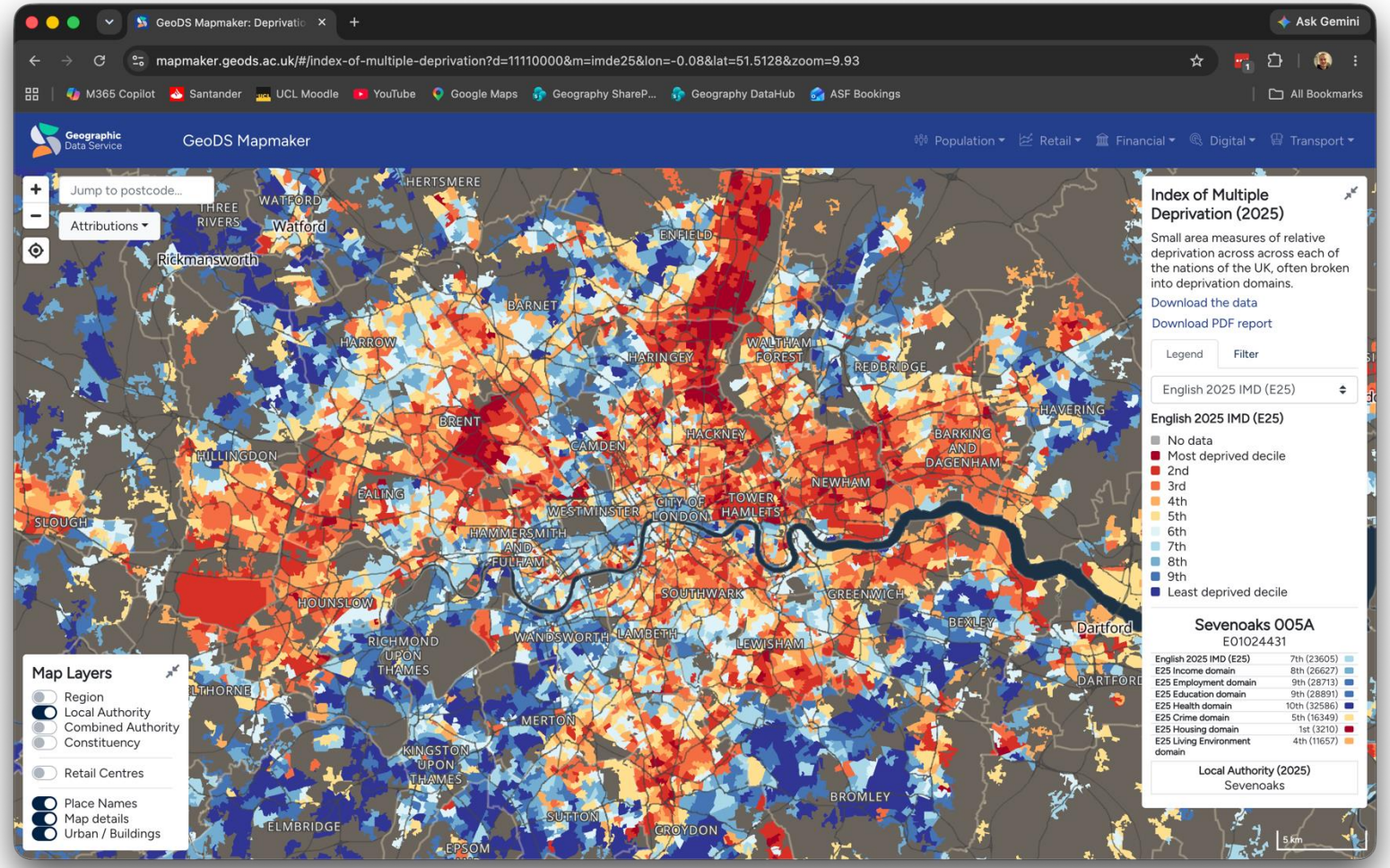
# Why is spatial data important?

- Many challenges and problems are inherently geographical.
- Almost all data is tied to specific locations and varies across space.
- Spatial data helps us answer the question: what is this place like compared to other places?
- Maps present spatial data in a format that the human brain understands more intuitively.

# Segregation in Cape Town



# Index of Multiple Deprivation in England

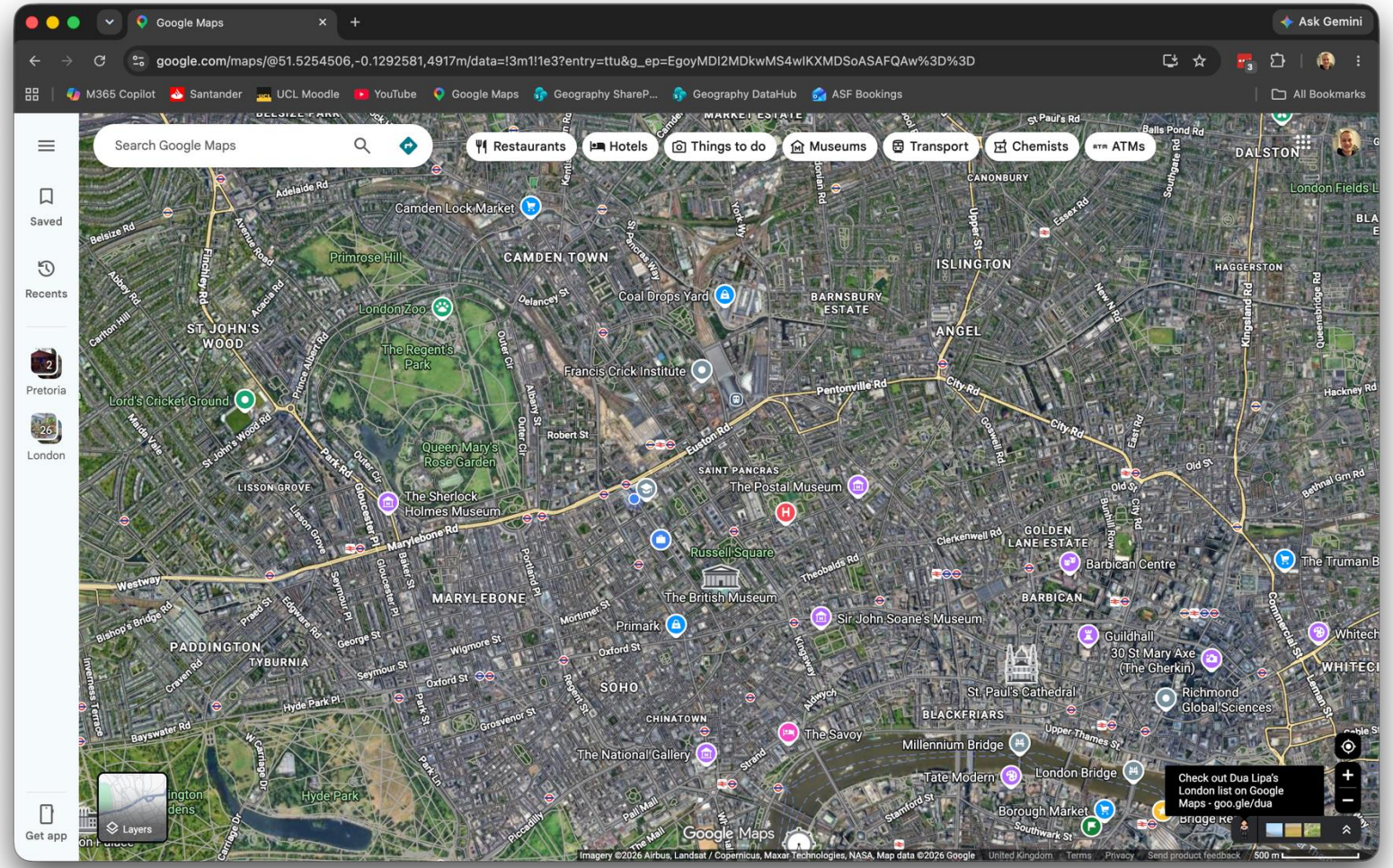


# GIScience and GISystems

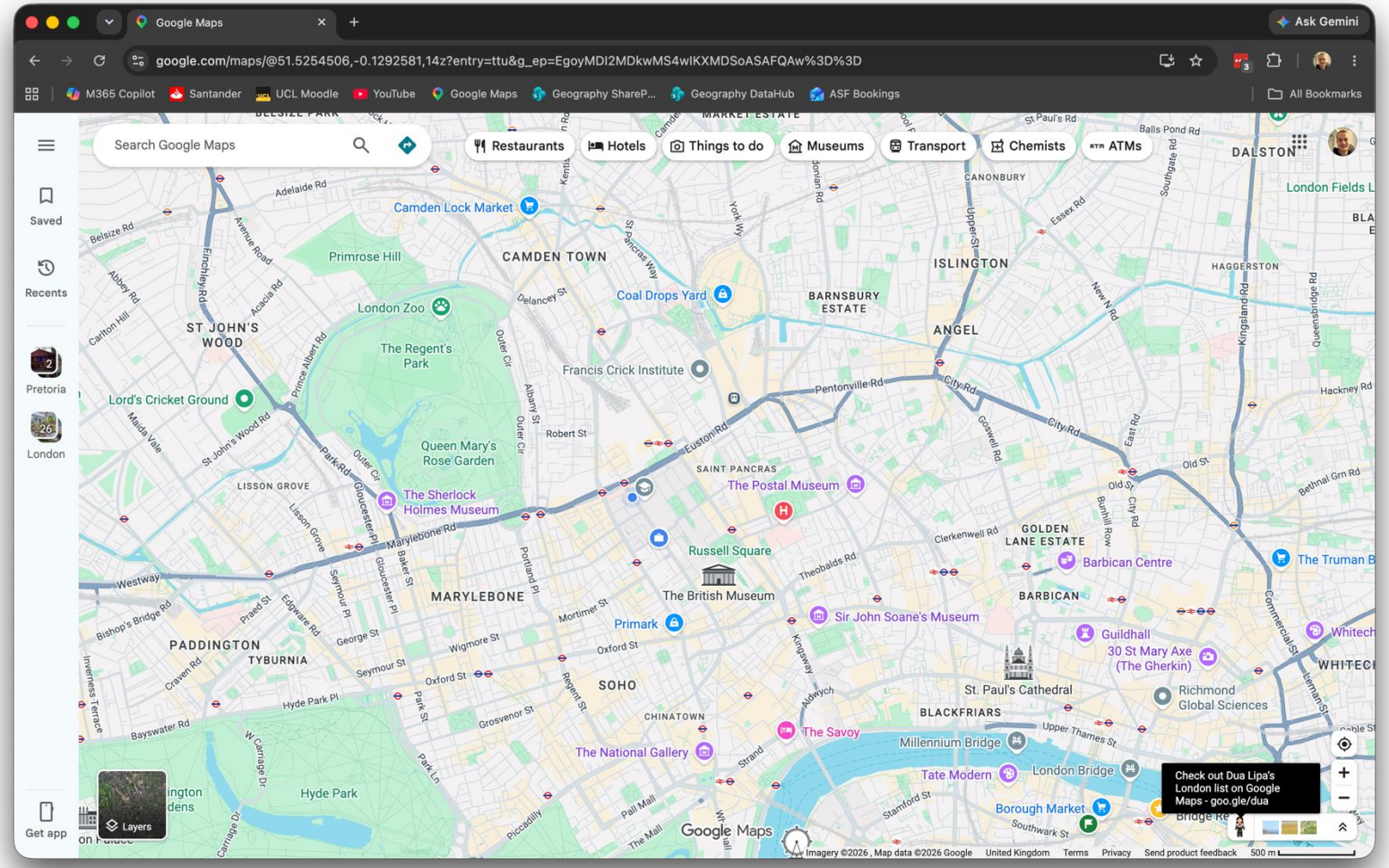
# Requirements

- GIScience: Theory
- GISystems: Methods and tools

# Digital representation



# Digital representation



# GIScience

- GIScience relies on representing spatial information in a digital format. Traditionally, geographic information is conveyed in two primary ways:

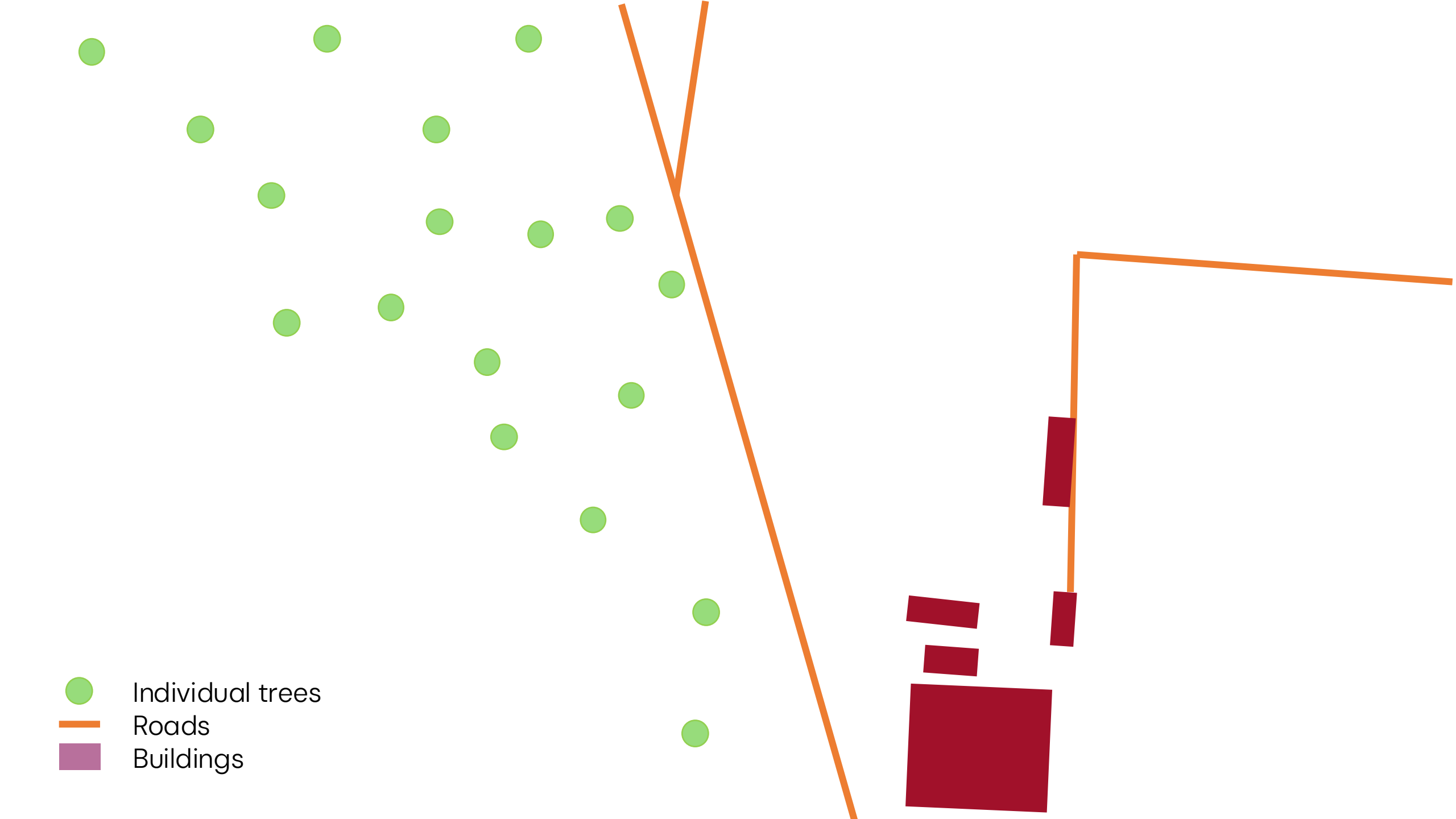
  Vector This method uses a finite set of discrete geometric objects, such as points, lines, and polygons, to represent spatial features.

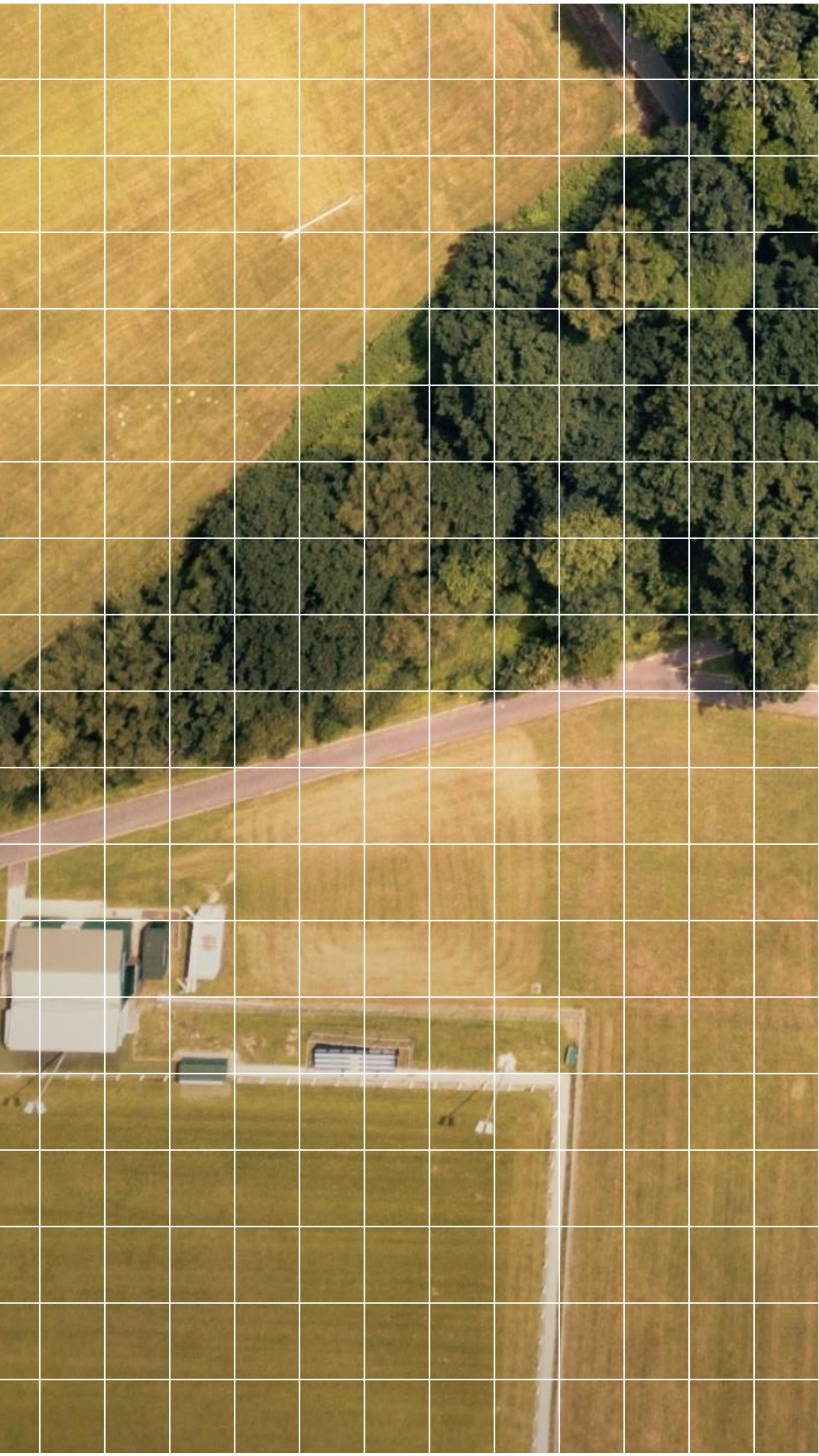
  Raster This approach employs images or grids to represent surfaces, with each cell or pixel holding a value, often indicating attributes like colours or measurements.
- Each feature also has attributes: a row in a table with information about it, such as a road name or a population count.







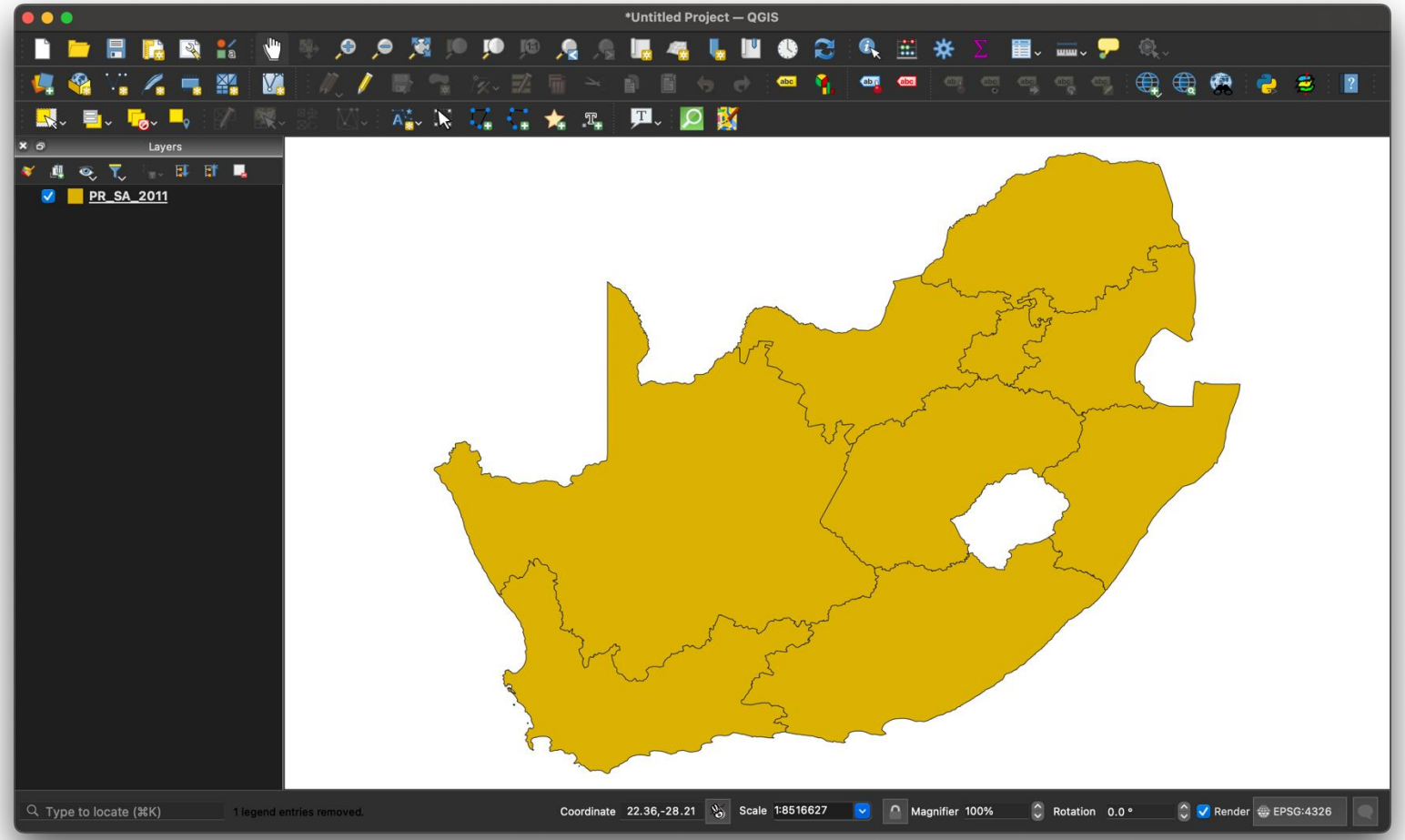

Individual trees
Roads
Buildings



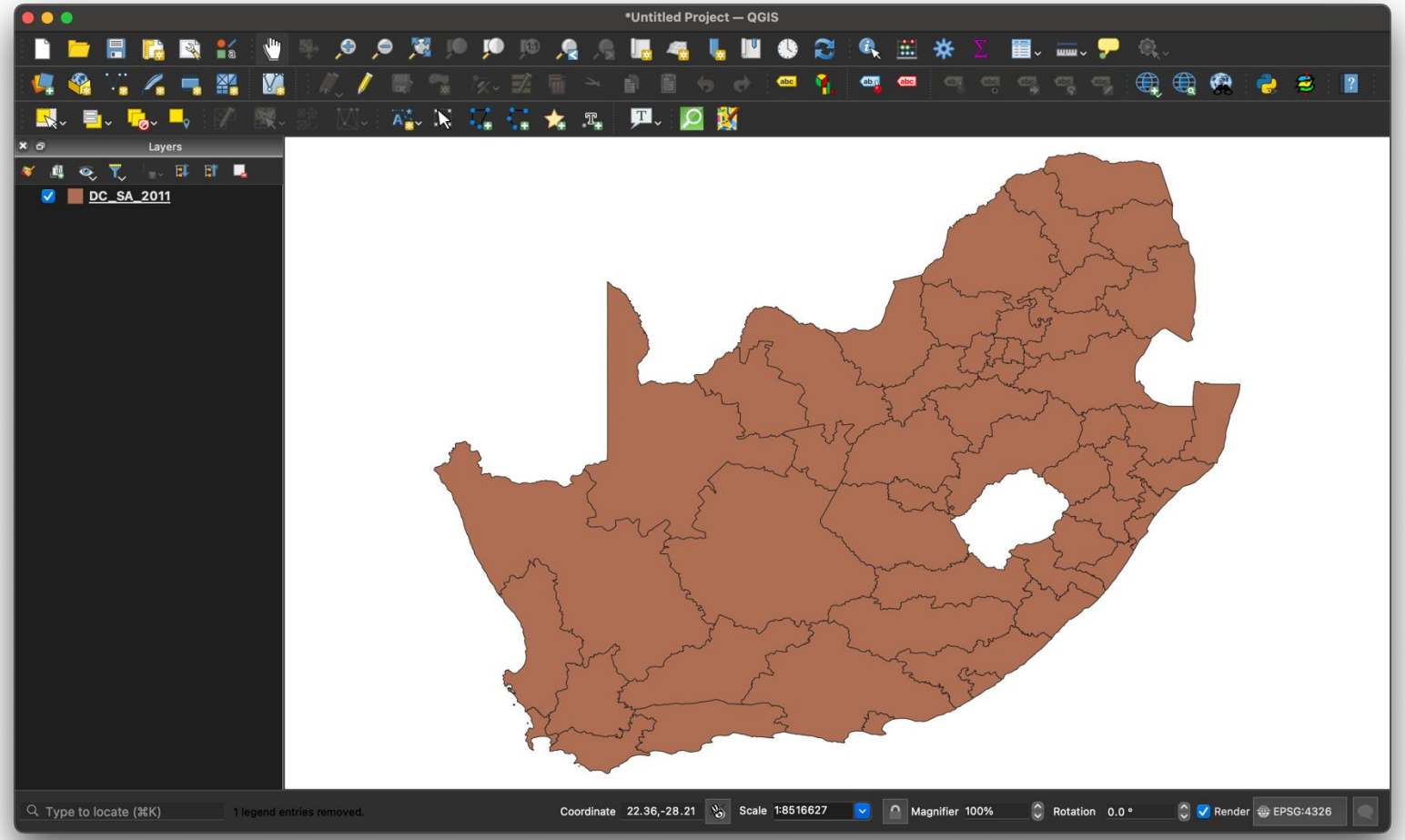


| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 8 | 9 | 9 | 10 | 0 | 10 | 10 | 10 | 0 | 0 | 0 | 7 | 5 | 3 | 0 | 0 | 0 | 0 | 1 |
| 8 | 9 | 9 | 10 | 10 | 0 | 10 | 9 | 9 | 0 | 0 | 5 | 3 | 0 | 0 | 0 | 0 | 0 | 0 |
| 8 | 8 | 9 | 9 | 10 | 0 | 0 | 9 | 8 | 7 | 5 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 5 | 8 | 8 | 9 | 10 | 10 | 0 | 9 | 7 | 5 | 0 | 0 | 5 | 5 | 5 | 0 | 0 | 0 | 1 |
| 3 | 5 | 8 | 9 | 9 | 10 | 0 | 0 | 3 | 0 | 0 | 0 | 5 | 0 | 0 | 1 | 0 | 0 | 2 |
| 2 | 5 | 8 | 8 | 9 | 9 | 10 | 0 | 0 | 0 | 1 | 5 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 2 | 4 | 6 | 8 | 8 | 9 | 0 | 0 | 0 | 1 | 5 | 0 | 0 | 5 | 5 | 5 | 0 | 0 | 1 |
| 0 | 3 | 6 | 8 | 8 | 0 | 0 | 0 | 0 | 5 | 0 | 5 | 5 | 5 | 5 | 5 | 0 | 0 | 0 |
| 2 | 2 | 5 | 8 | 0 | 0 | 0 | 0 | 0 | 0 | 5 | 5 | 5 | 5 | 5 | 5 | 3 | 0 | 0 |
| 0 | 2 | 5 | 0 | 0 | 1 | 2 | 3 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 5 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 5 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 2 | 2 | 2 | 2 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 4 | 0 | 3 | 0 |
| 1 | 1 | 1 | 1 | 2 | 2 | 3 | 3 | 3 | 3 | 1 | 1 | 1 | 1 | 1 | 2 | 3 | 4 | 3 |

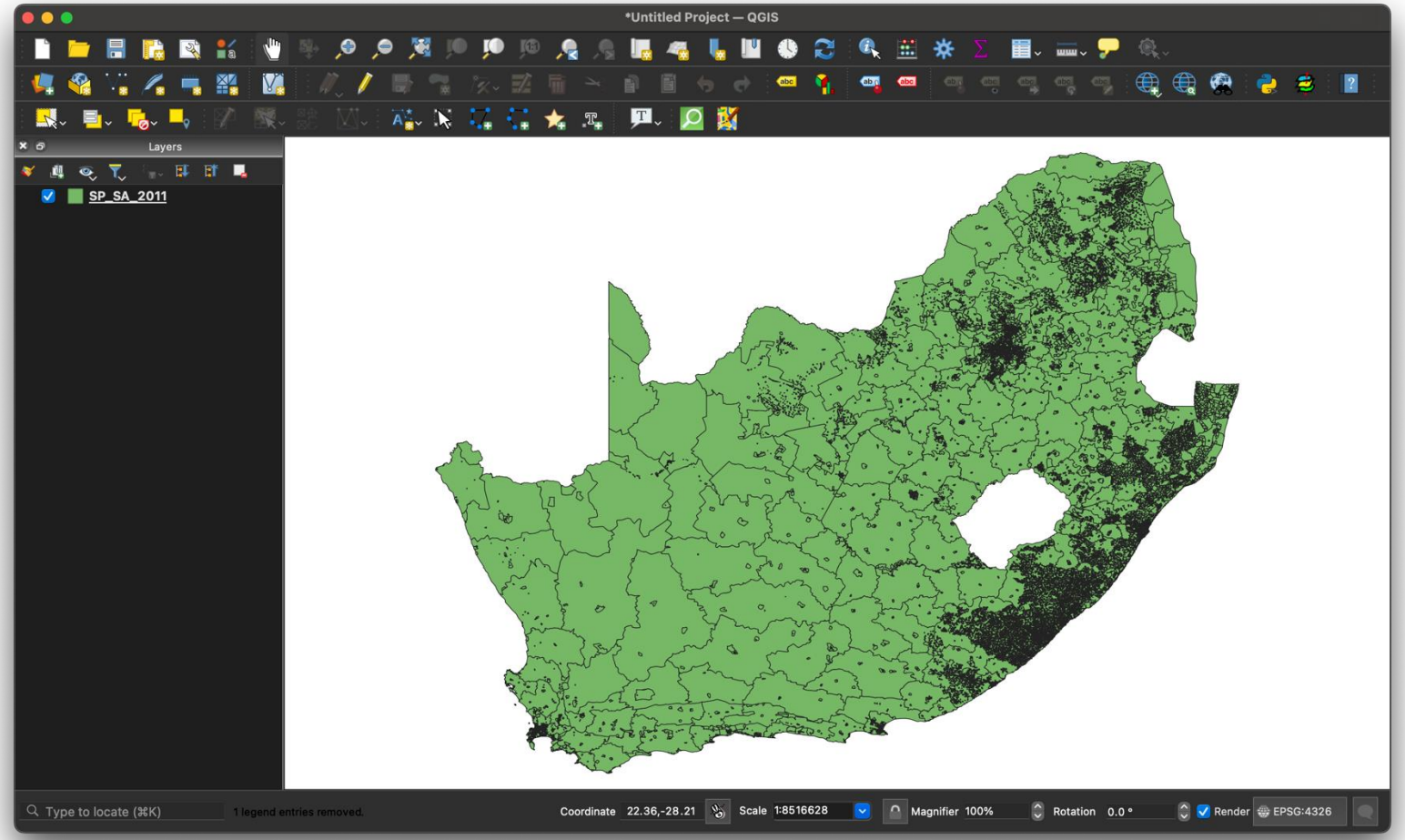
# Same place, different levels of detail



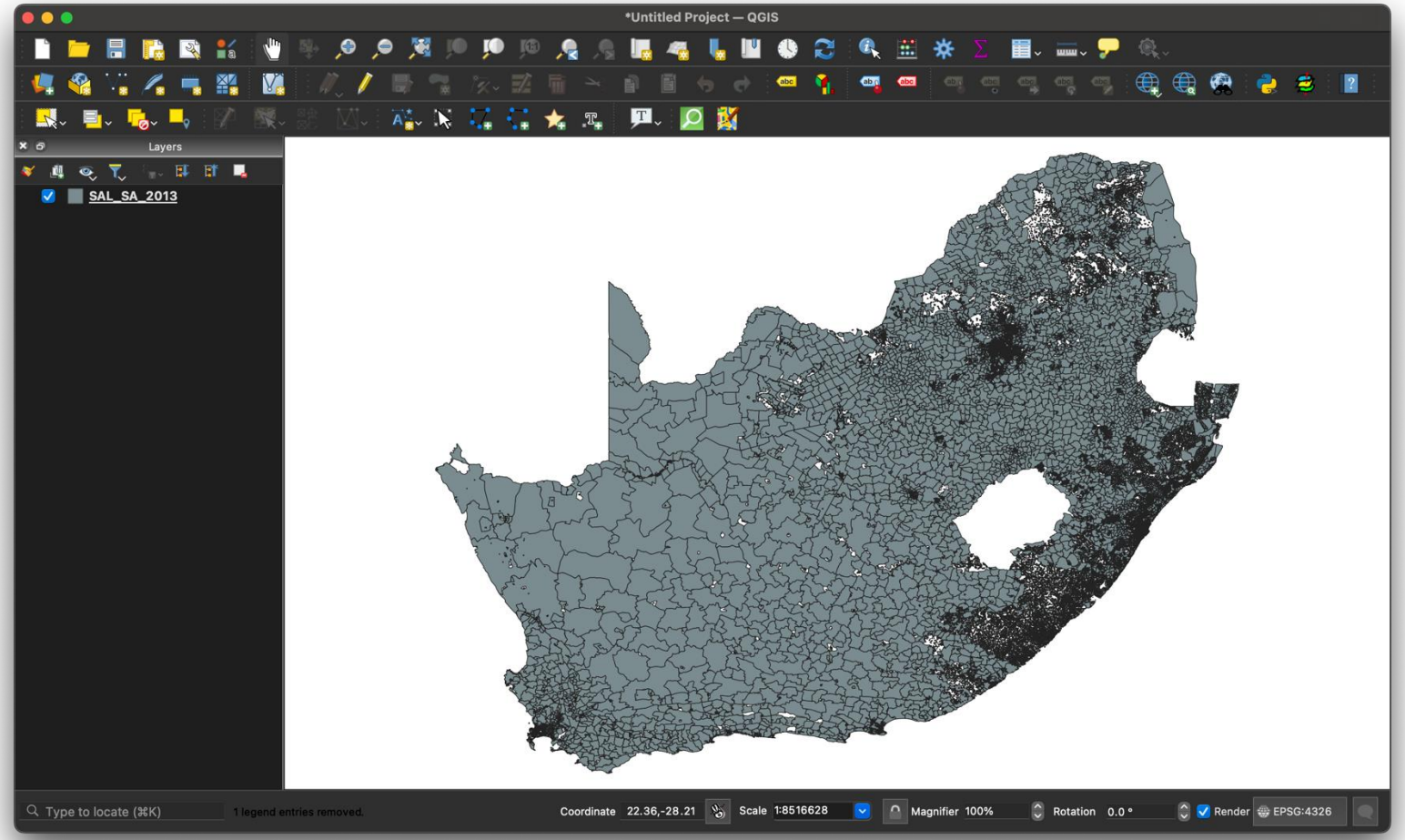
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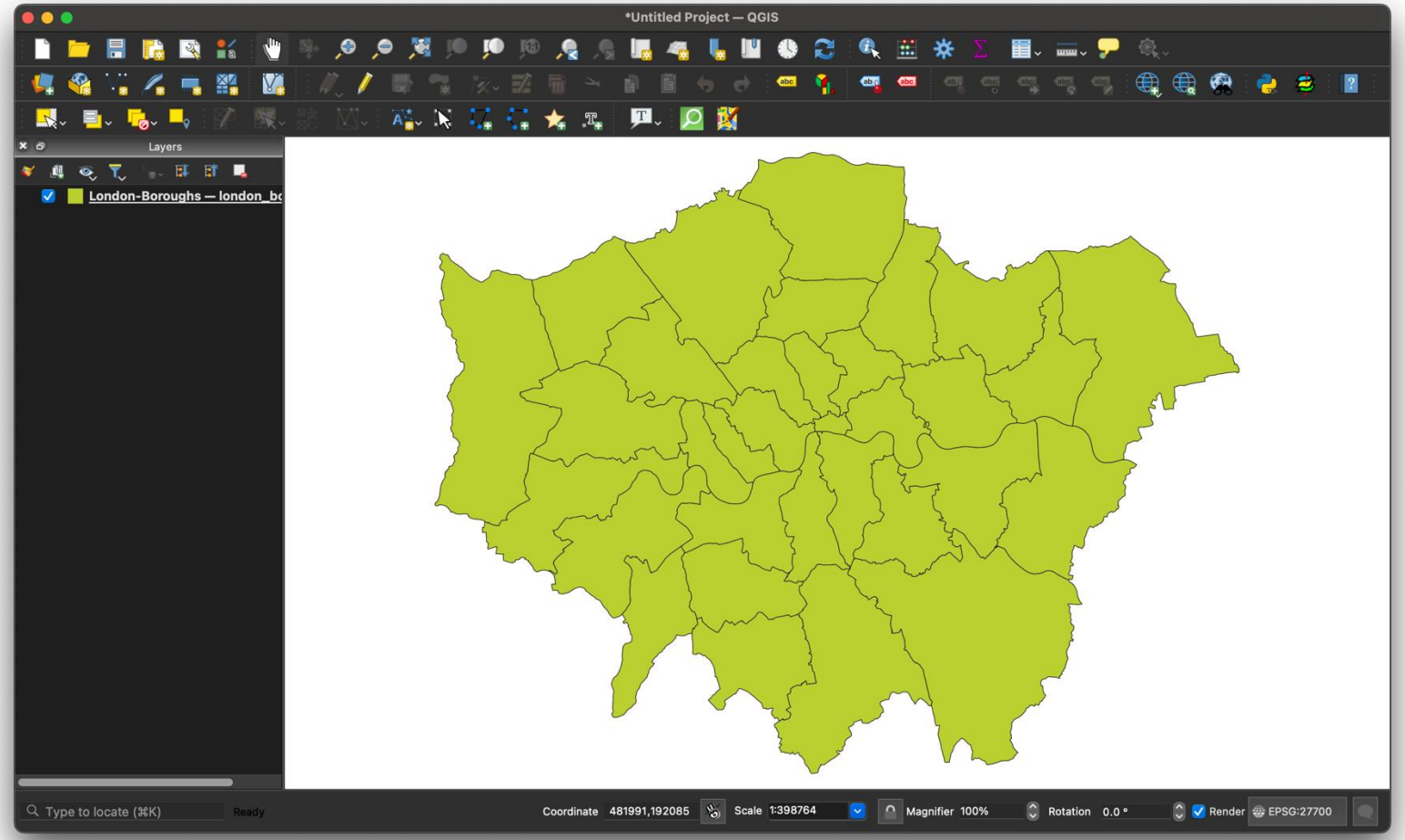
# Same place, different levels of detail



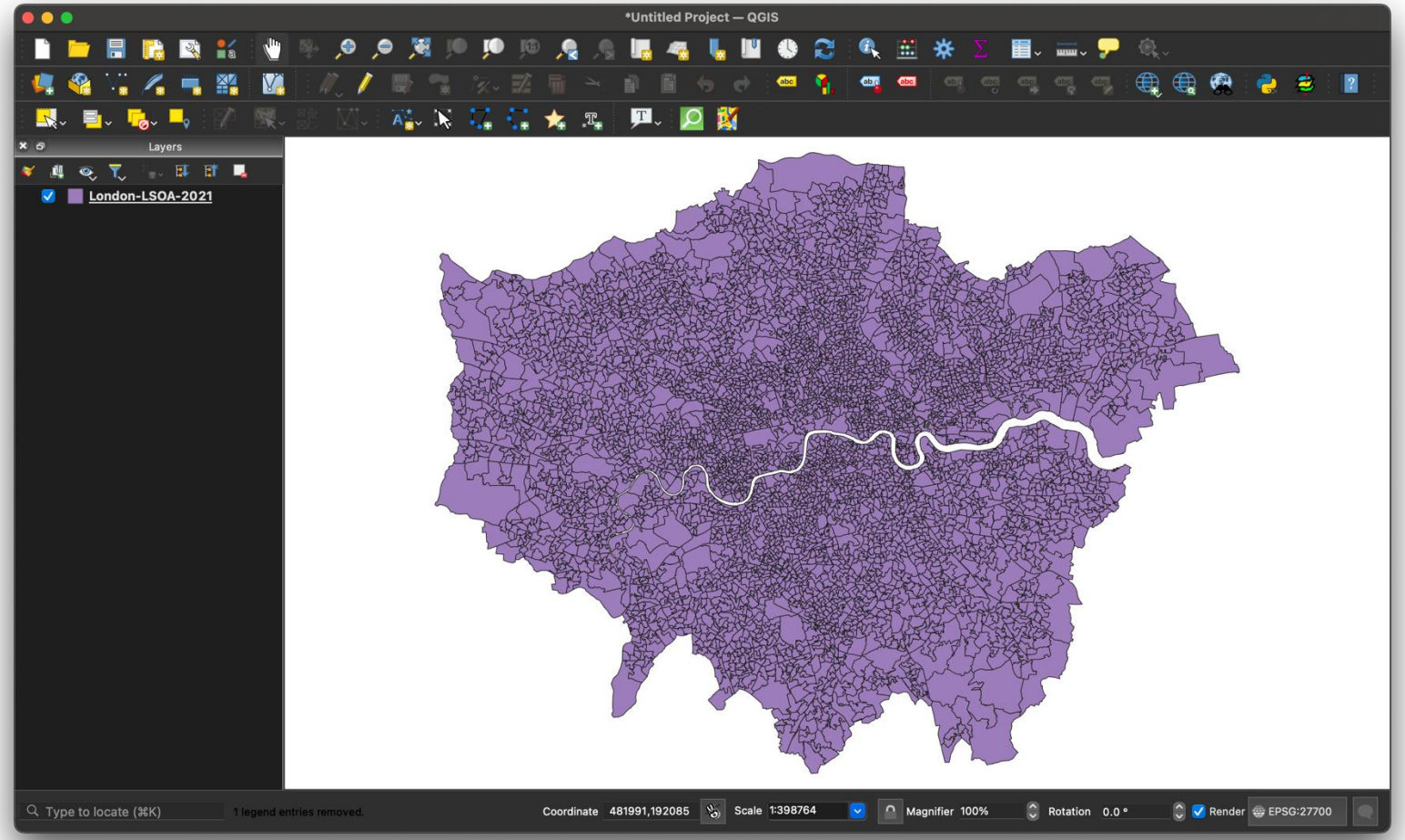
# Same place, different levels of detail



# Same place, different levels of detail



# Same place, different levels of detail




Lower layer Super Output Areas (LSOAs): 1,000 to 3,000 residents each.

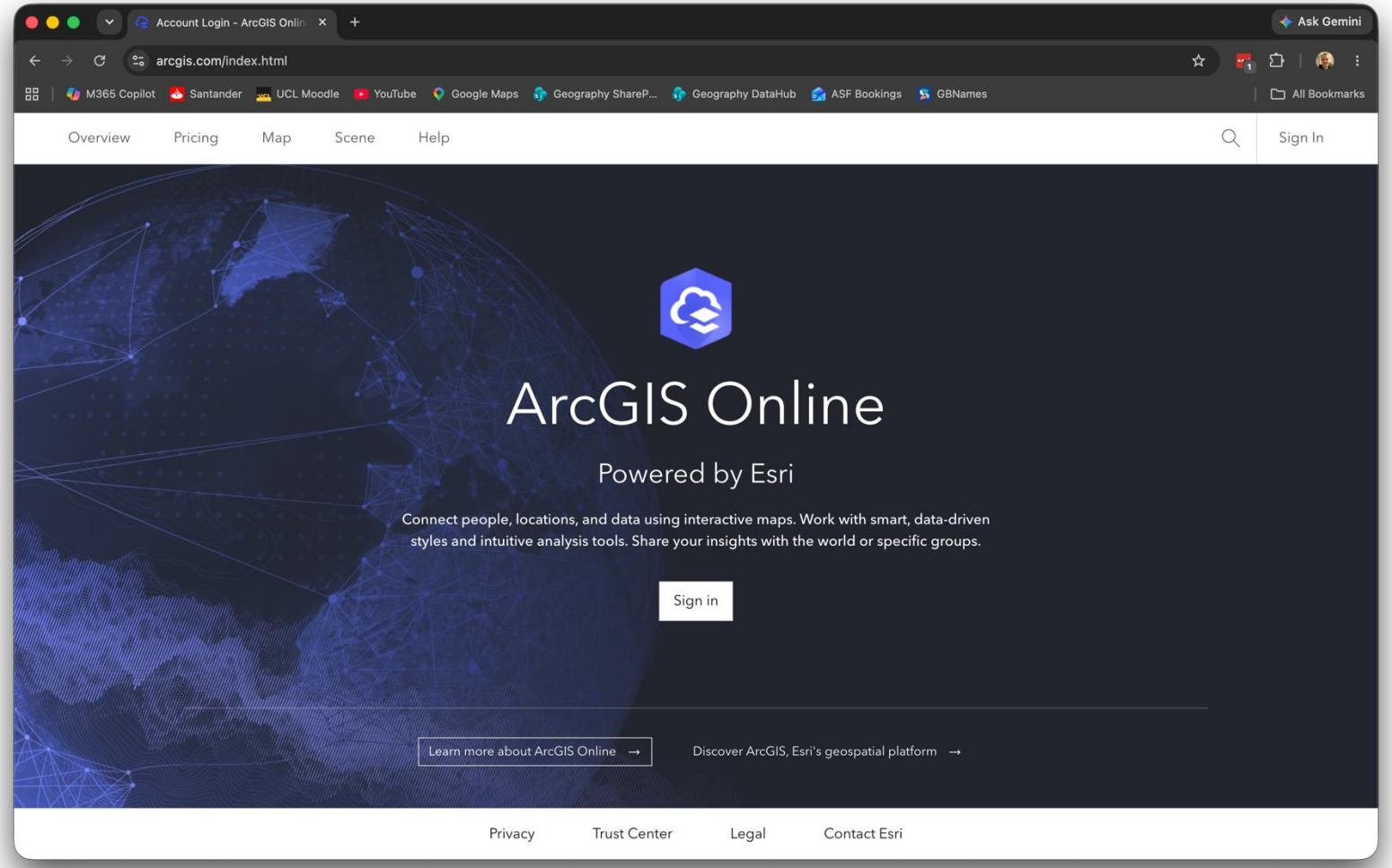
# Father of GIS

- Roger F. Tomlinson (1933–2014)
- Ph.D. dissertation: "*The application of electronic computing methods and techniques to the storage, compilation, and assessment of mapped data*" (1974, UCL).
- Conceived the idea of analysing multiple layers of spatial data within a single environment as well as how to represent such spatial data in a digital format.

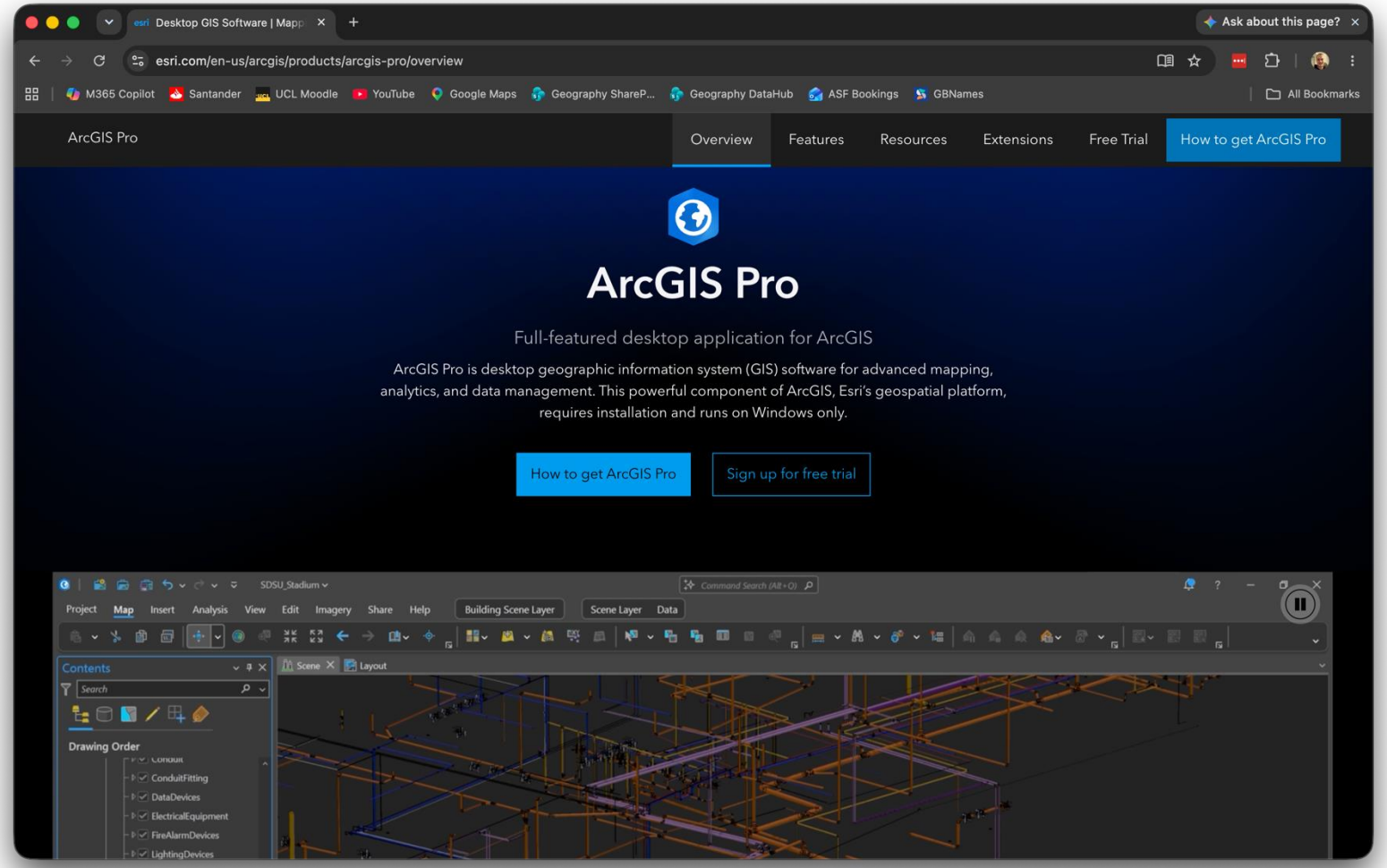
# Geographic Information Systems (GIS)

- To effectively handle spatial information, we need tools that can:
  - Collect spatial data
  - Store spatial data
  - Analyse spatial data
  - Present spatial data
- Geographic Information Systems help us to manage spatial data.

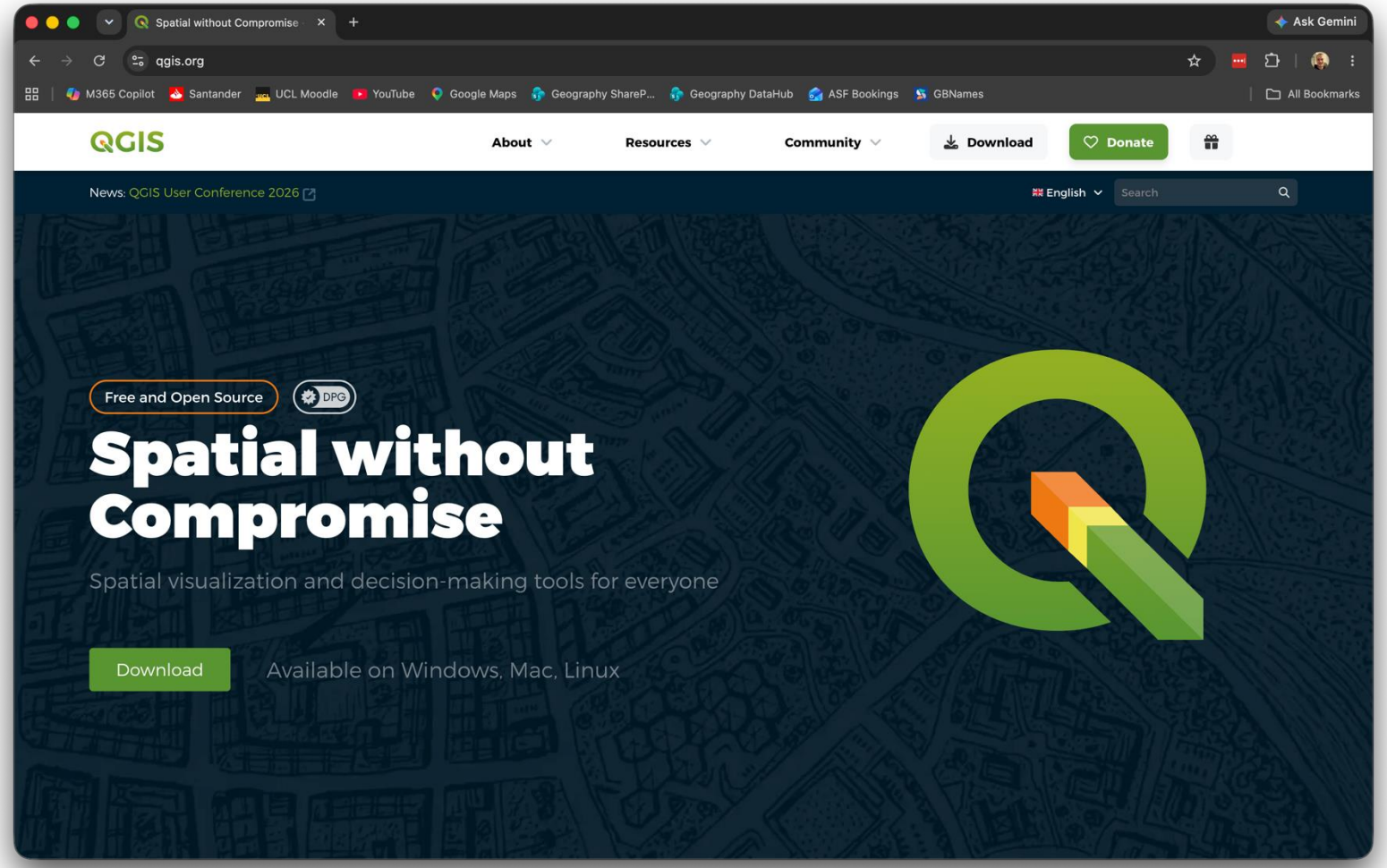
# Geographic Information Systems



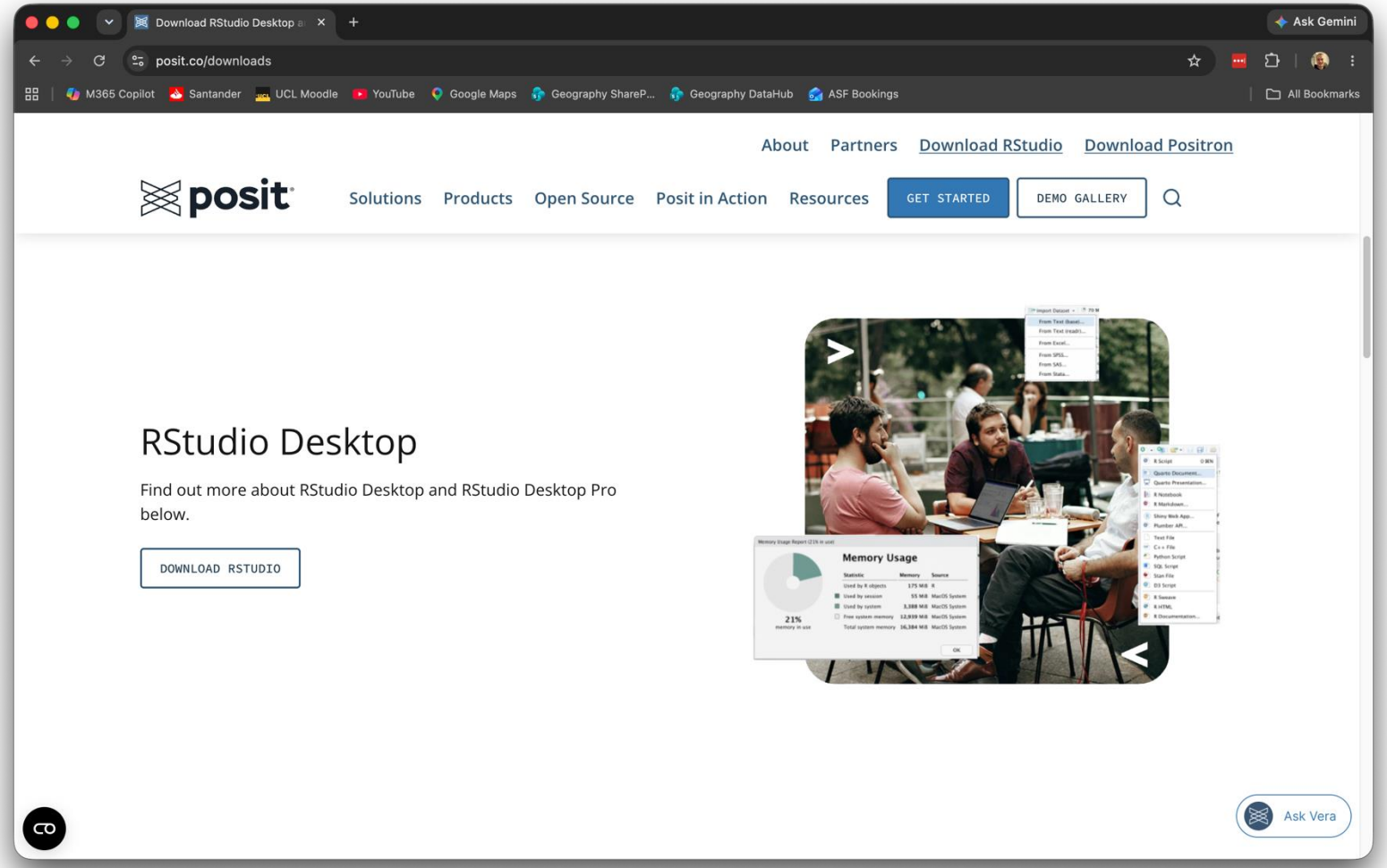
# Geographic Information Systems



# Geographic Information Systems



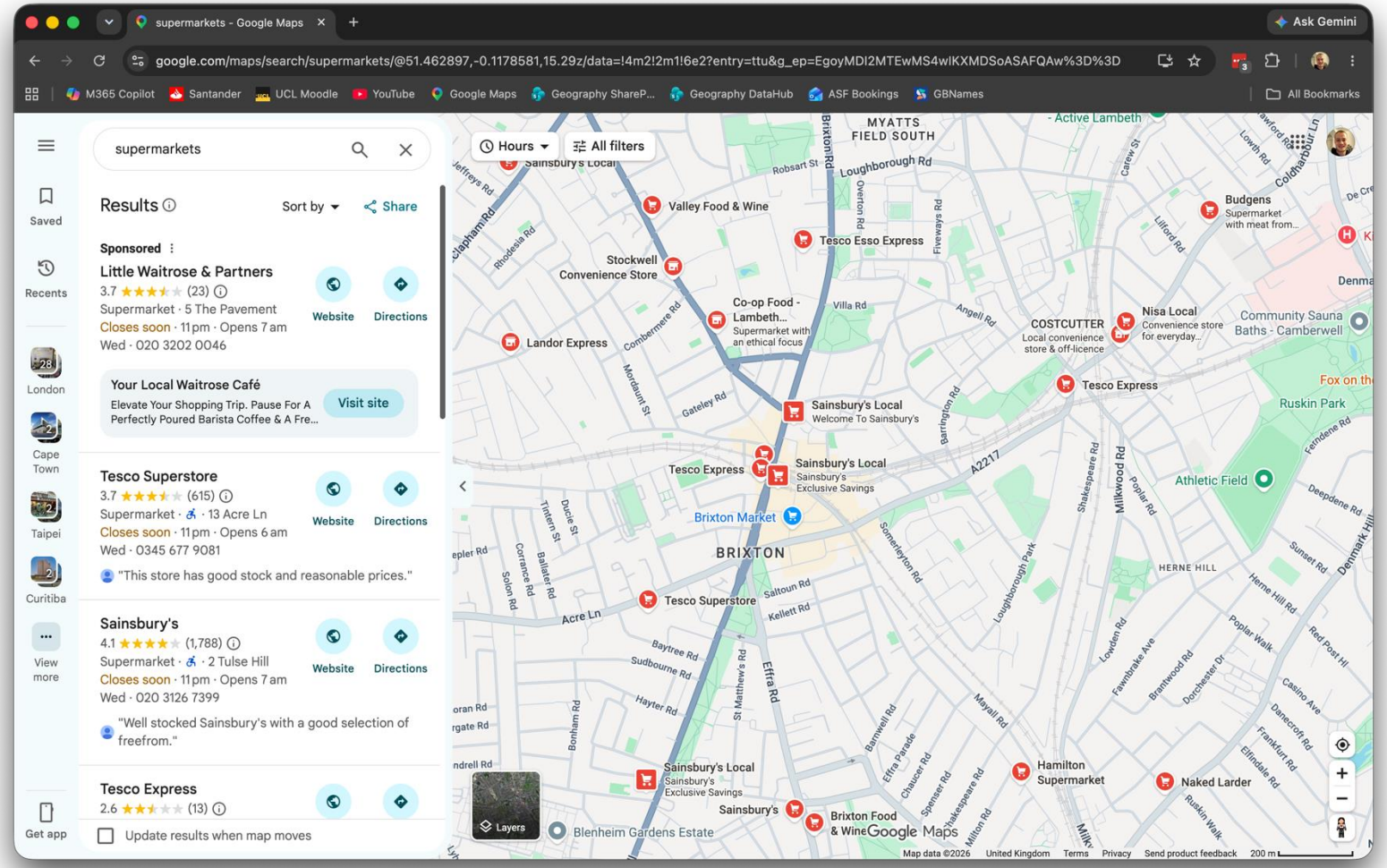
# Geographic Information Systems



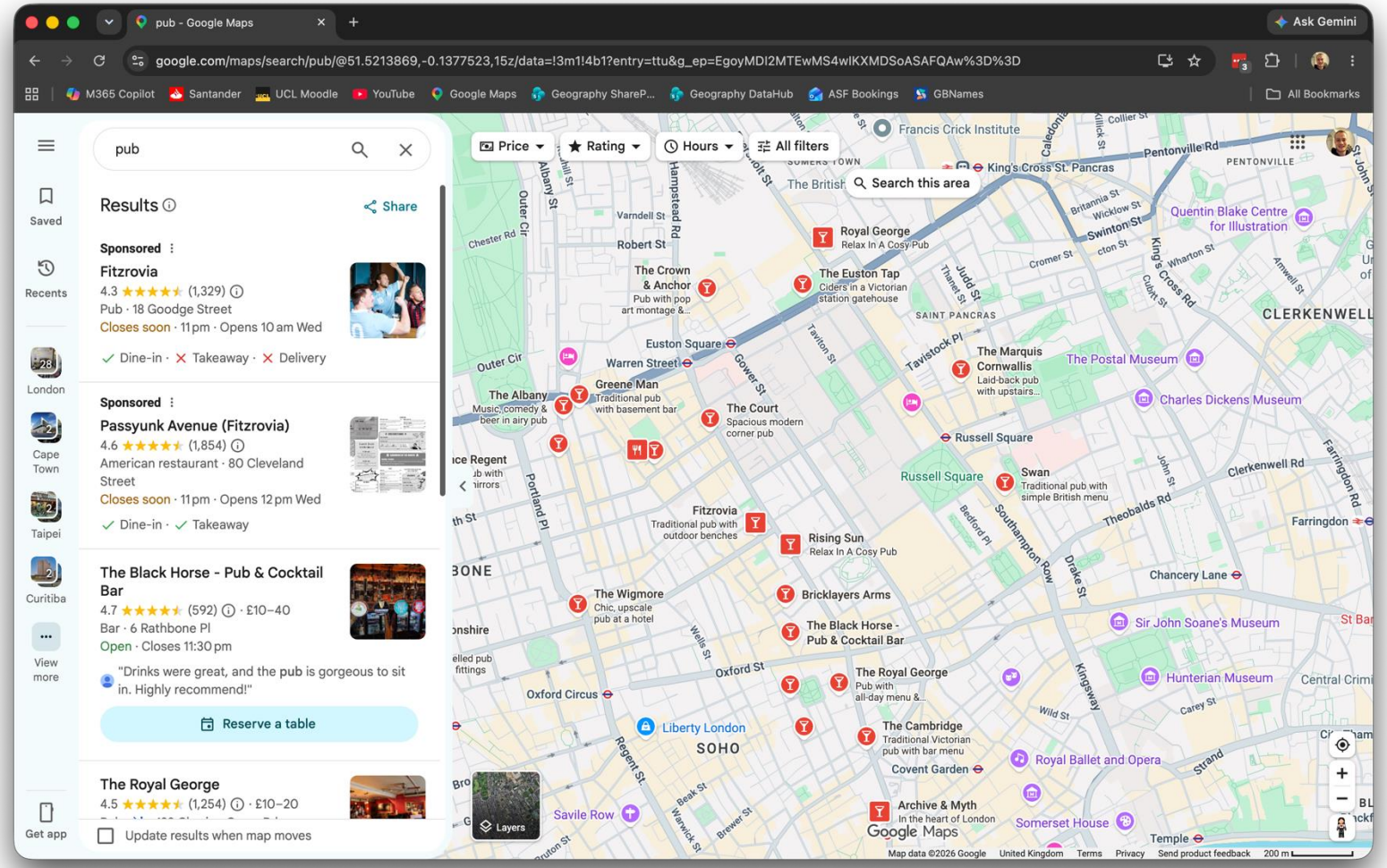
# Spatial analysis

- We can use GIScience and GISystems to convert spatial data into actionable information by quantifying elements such as distributions and spatial processes.
- In plain words: finding patterns in where things are and how places compare.

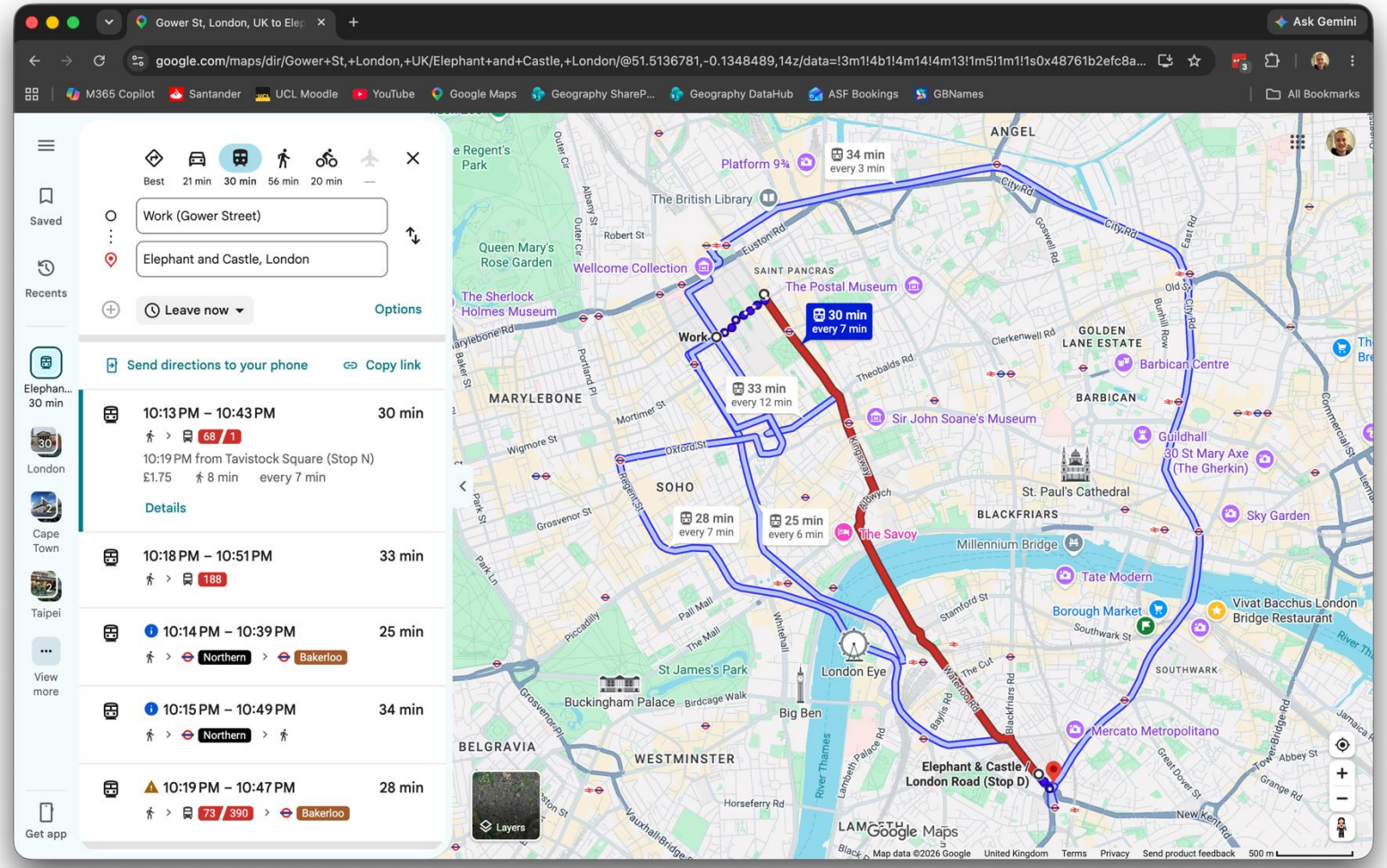
# Spatial analysis: finding locations



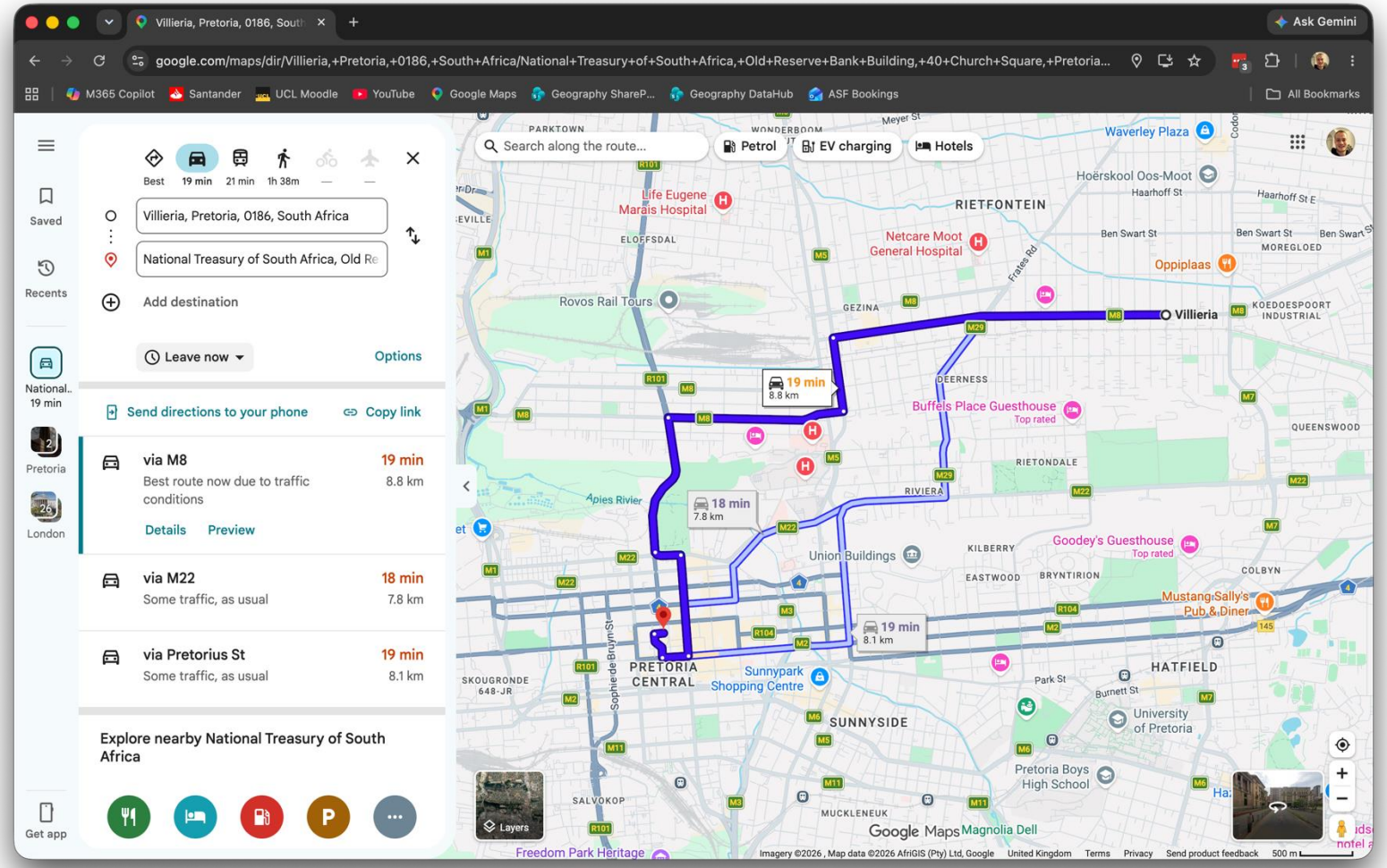
# Spatial analysis: finding locations



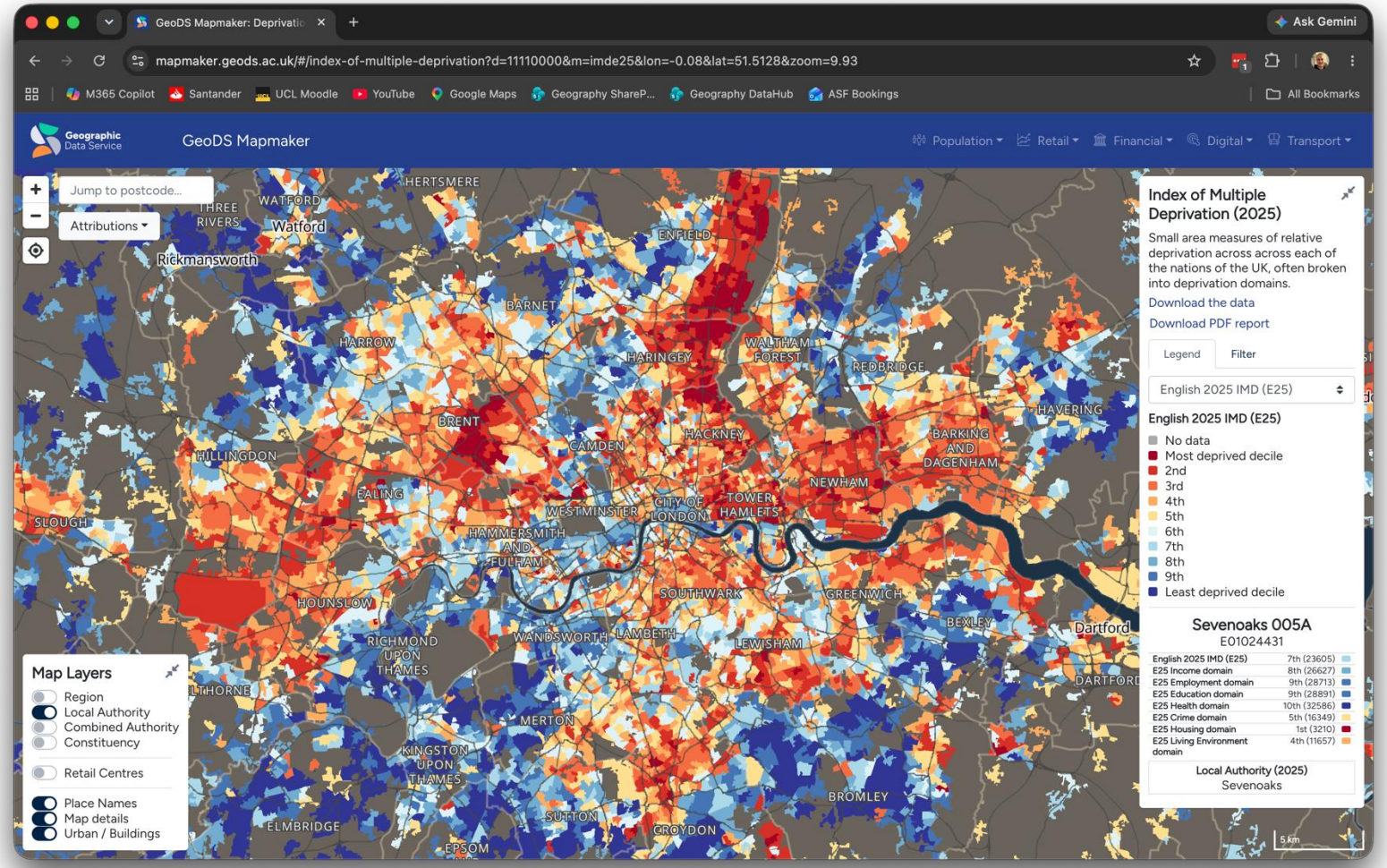
# Spatial analysis: finding routes



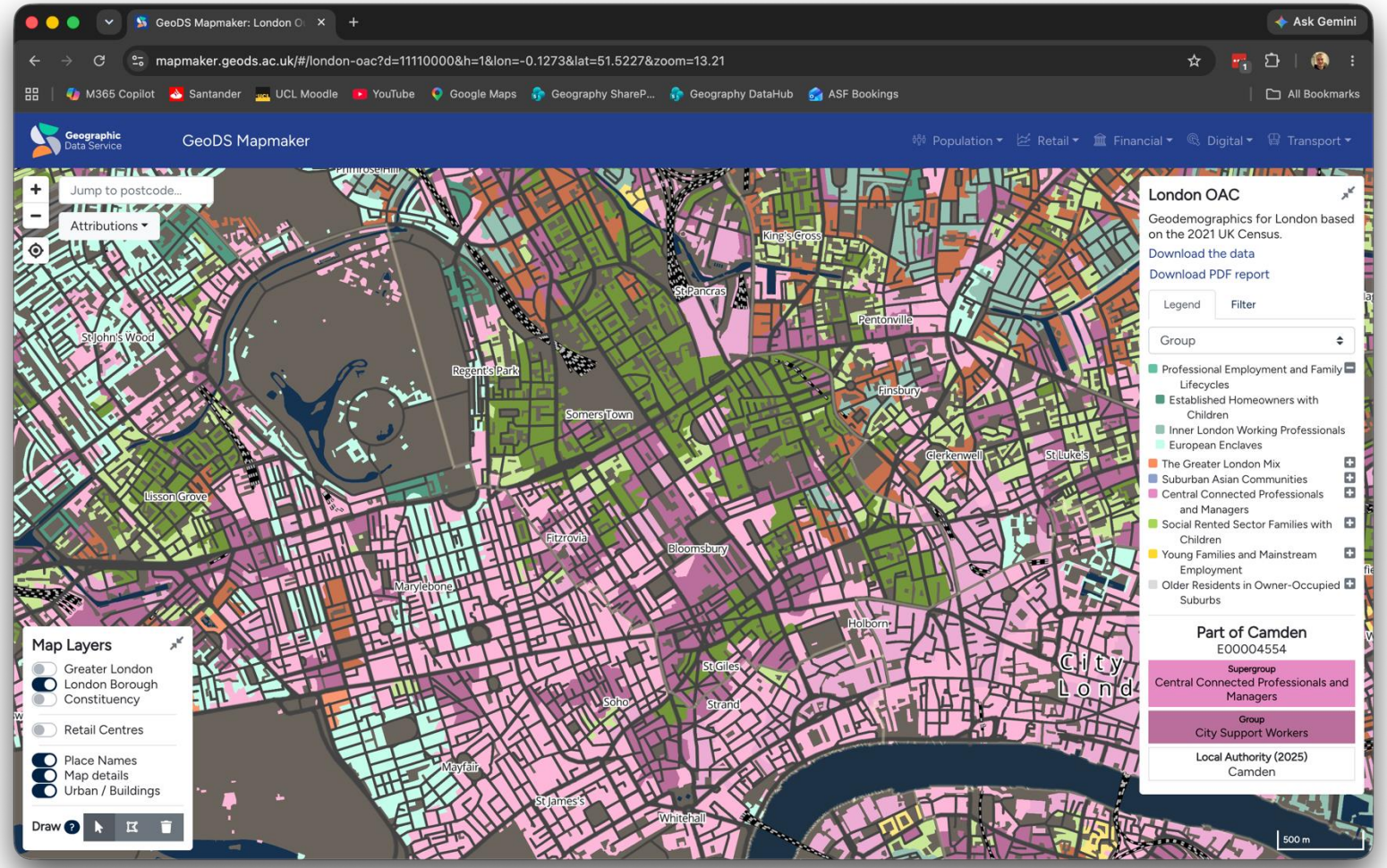
# Spatial analysis: finding routes



# Spatial analysis: comparing places



# Spatial analysis: comparing places



# File formats: Geopackage

- A GeoPackage is an open, standards-based, platform-independent, portable, self-describing, compact format for transferring geospatial data.
- It stores spatial data layers in a single file, based on an SQLite database.
- How to spot in the wild: .gpkg

# File formats: Shapefile

- `.shp` Contains the feature geometry.
- `.shx` Index file which stores the position of the feature's ID in the `.shp` file.
- `.dbf` Stores all attribute information associated with the records.
- `.prj` Contains the coordinate system information and projection.
- `.xml` General metadata.
- `.cpg` Encoding information.
- `.sbn` *O*ptimisation file for spatial queries.

# File formats: Shapefile



# Coordinate reference systems

- A coordinate reference system (CRS) defines how coordinates relate to real places on the Earth.
- British National Grid (EPSG:27700): metres, used for Great Britain.
- WGS 84 (EPSG:4326): latitude and longitude in degrees, used by GPS.
- Layers in different CRSs can end up in the wrong place.

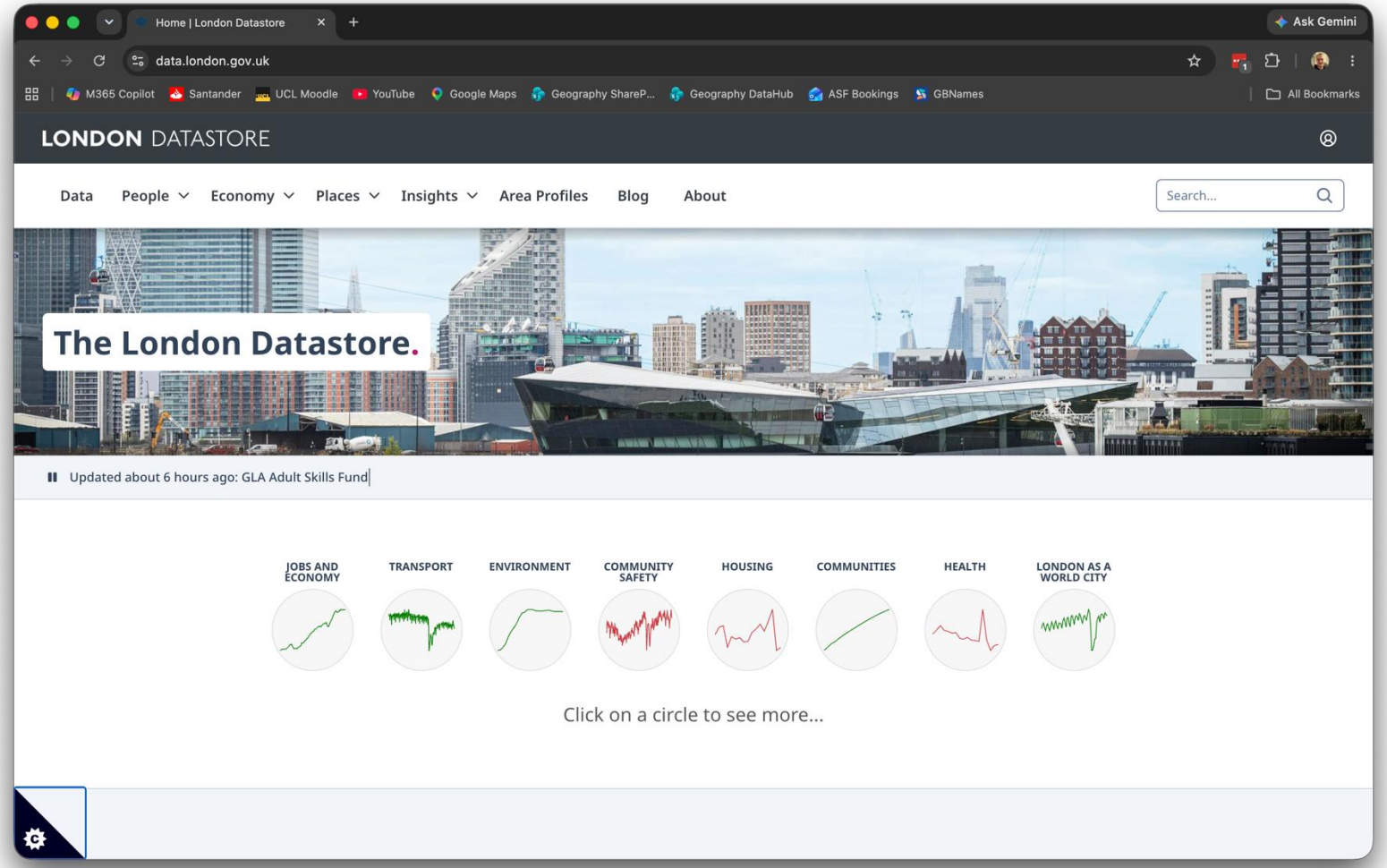
# Mentimeter

- Go to www.menti.com.
- Use code: XXXX XXXX

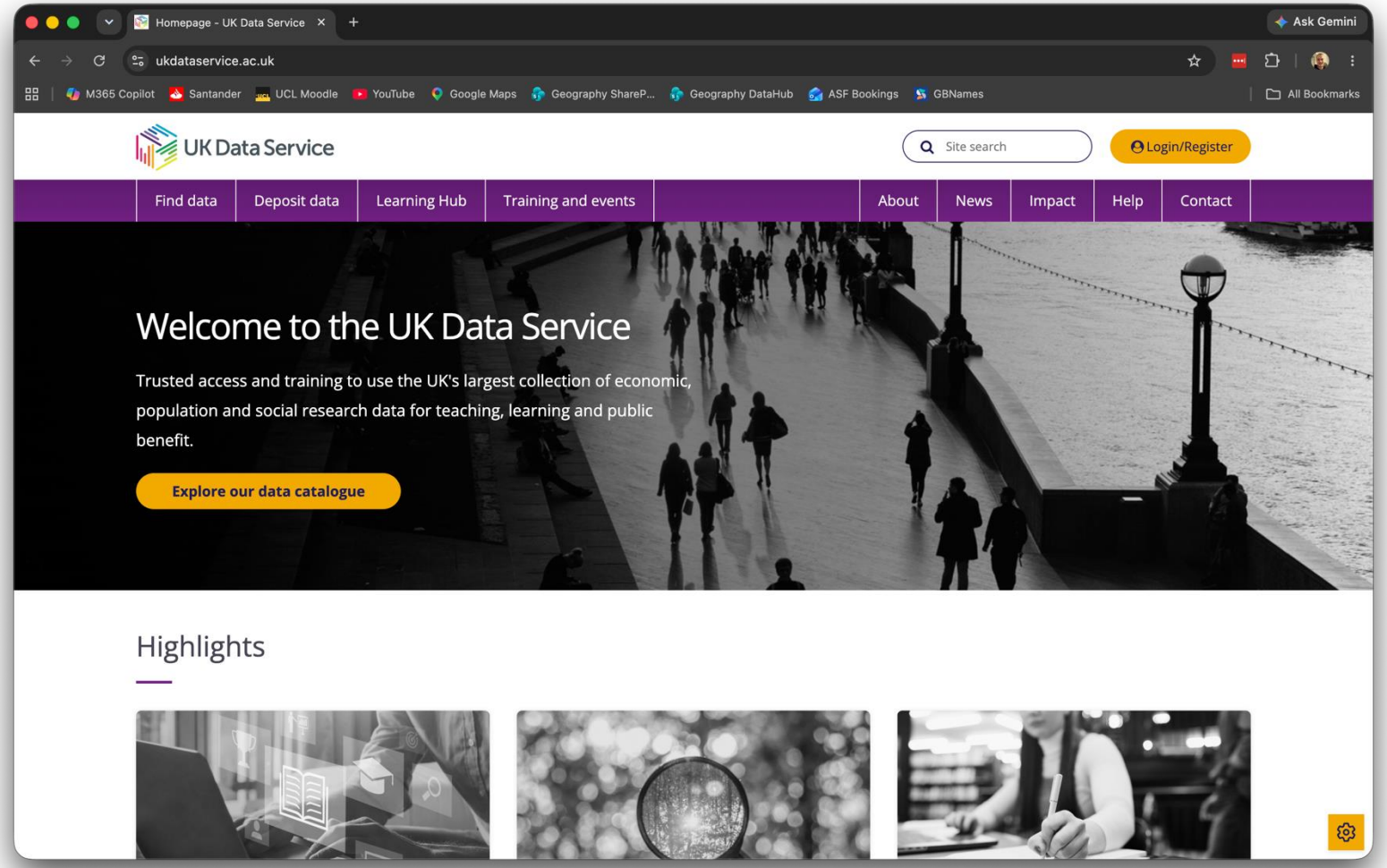
# Break

# Data sources

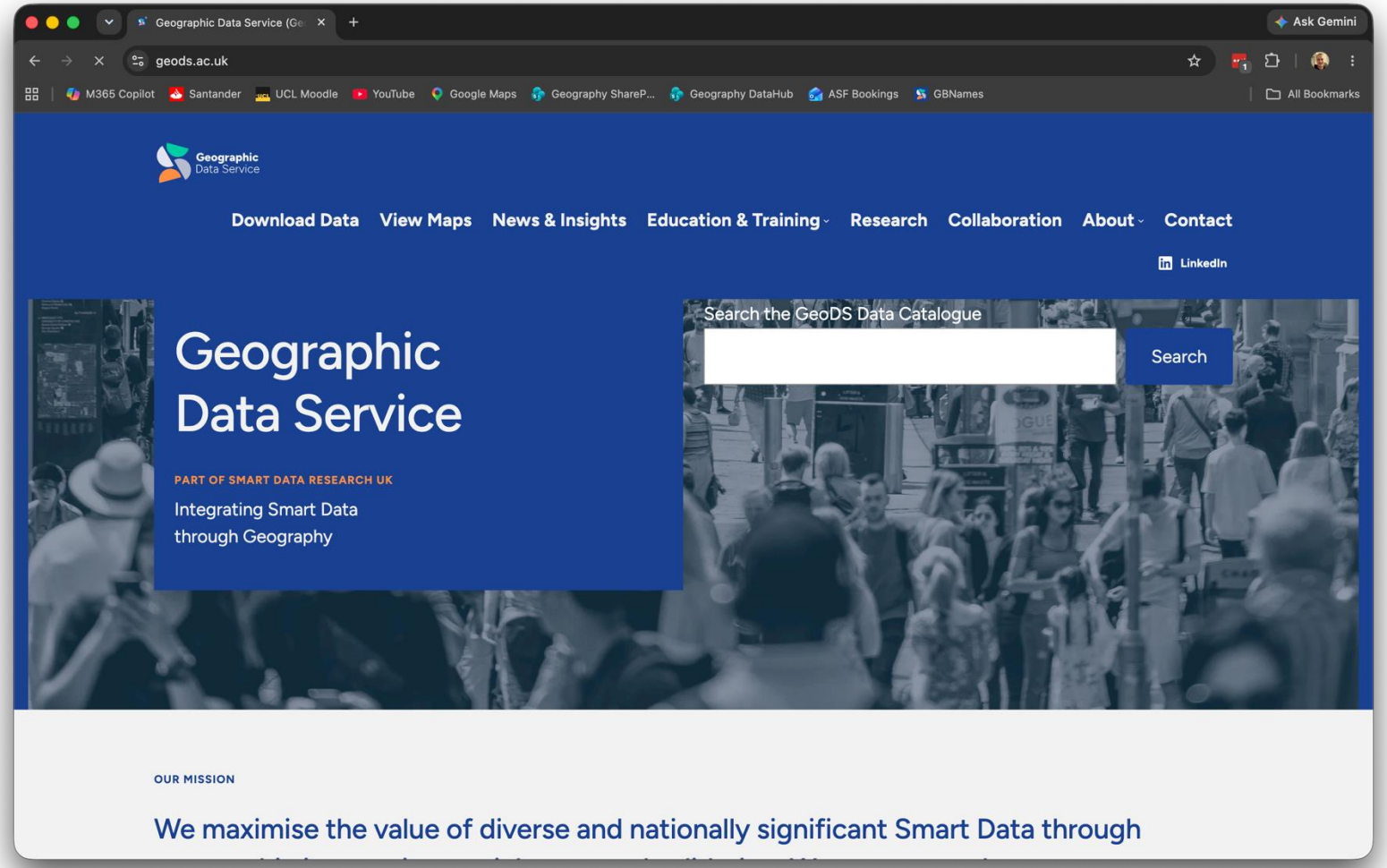
# London Datastore



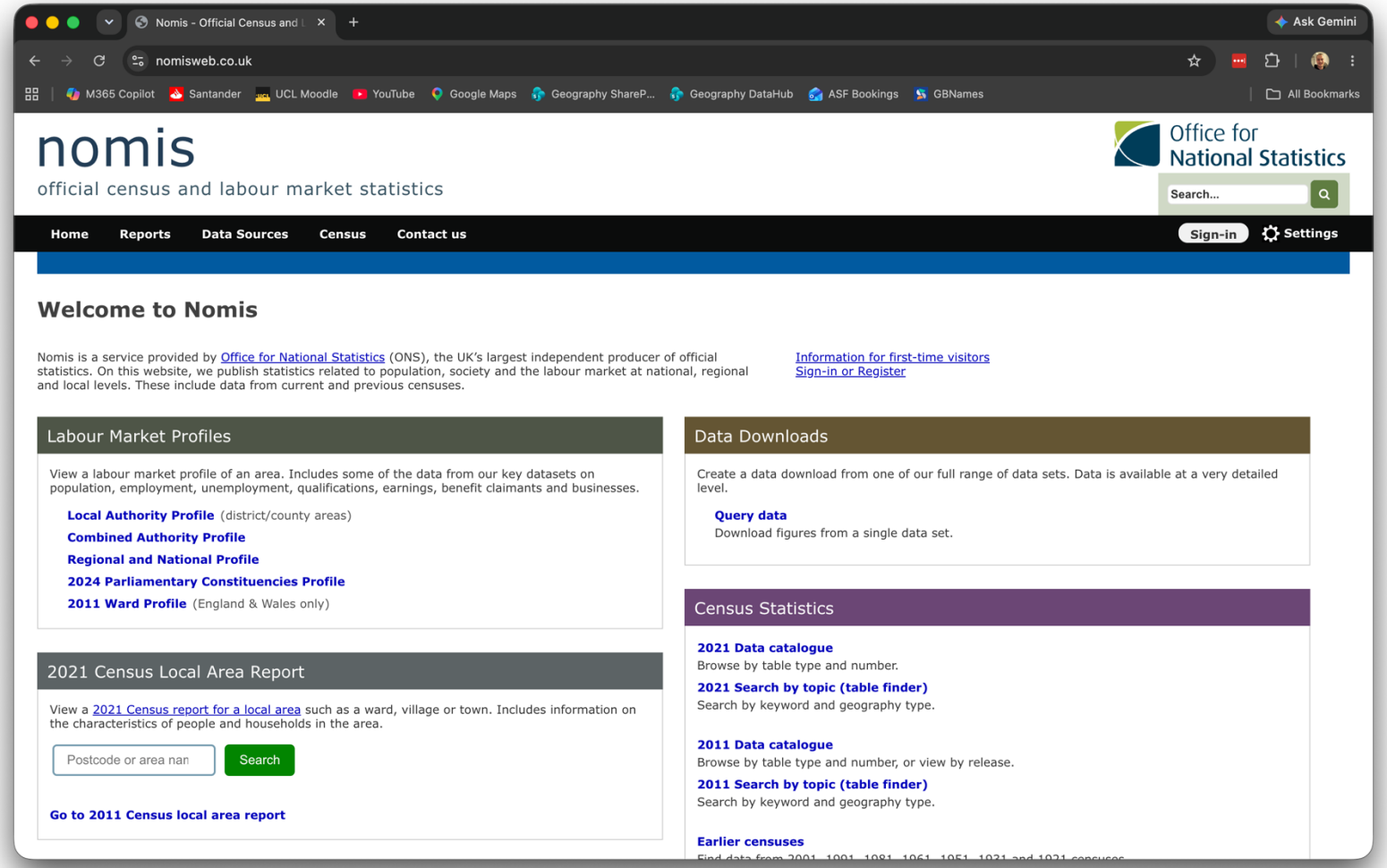
# UK Data Service



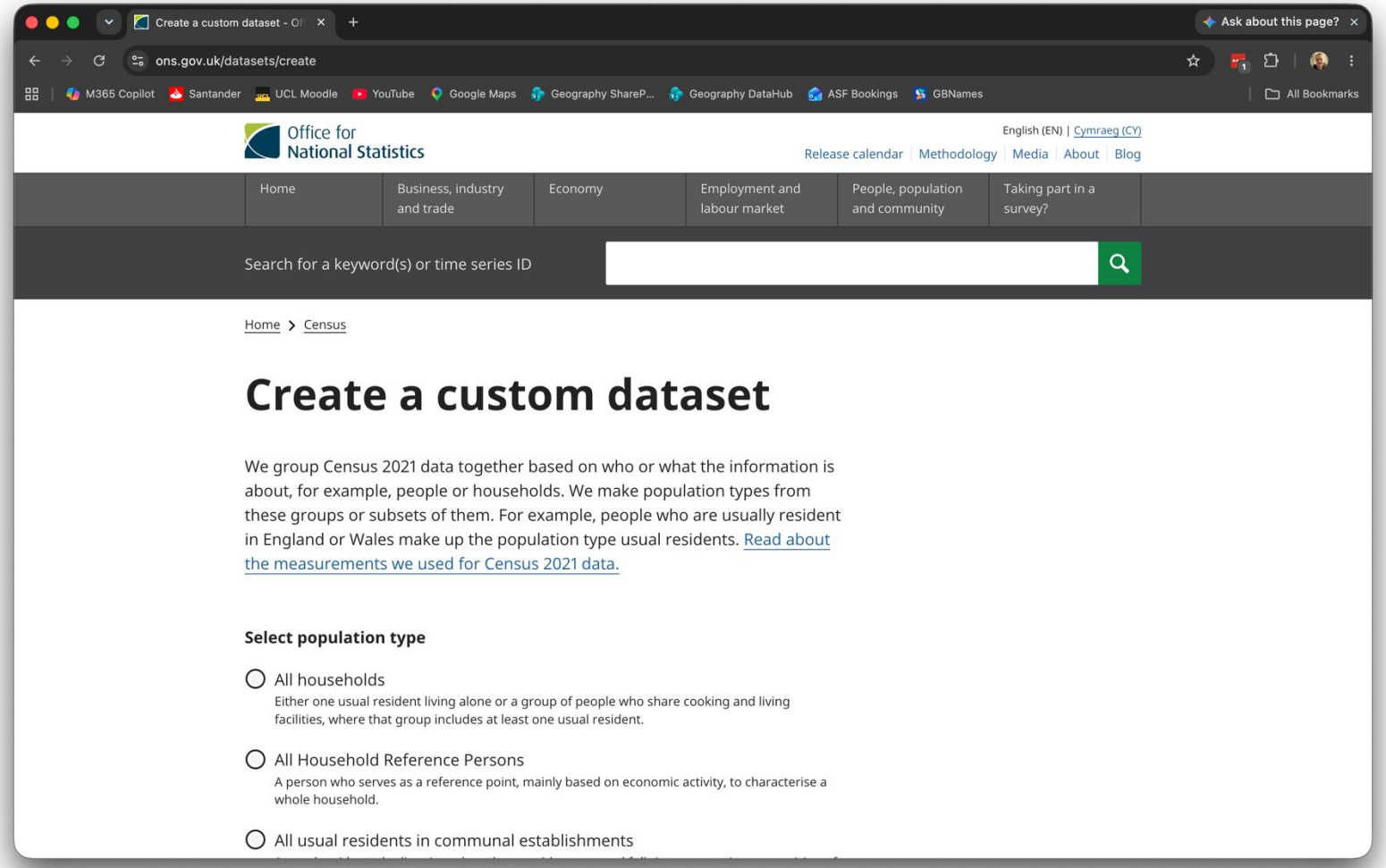
# Geographic Data Service



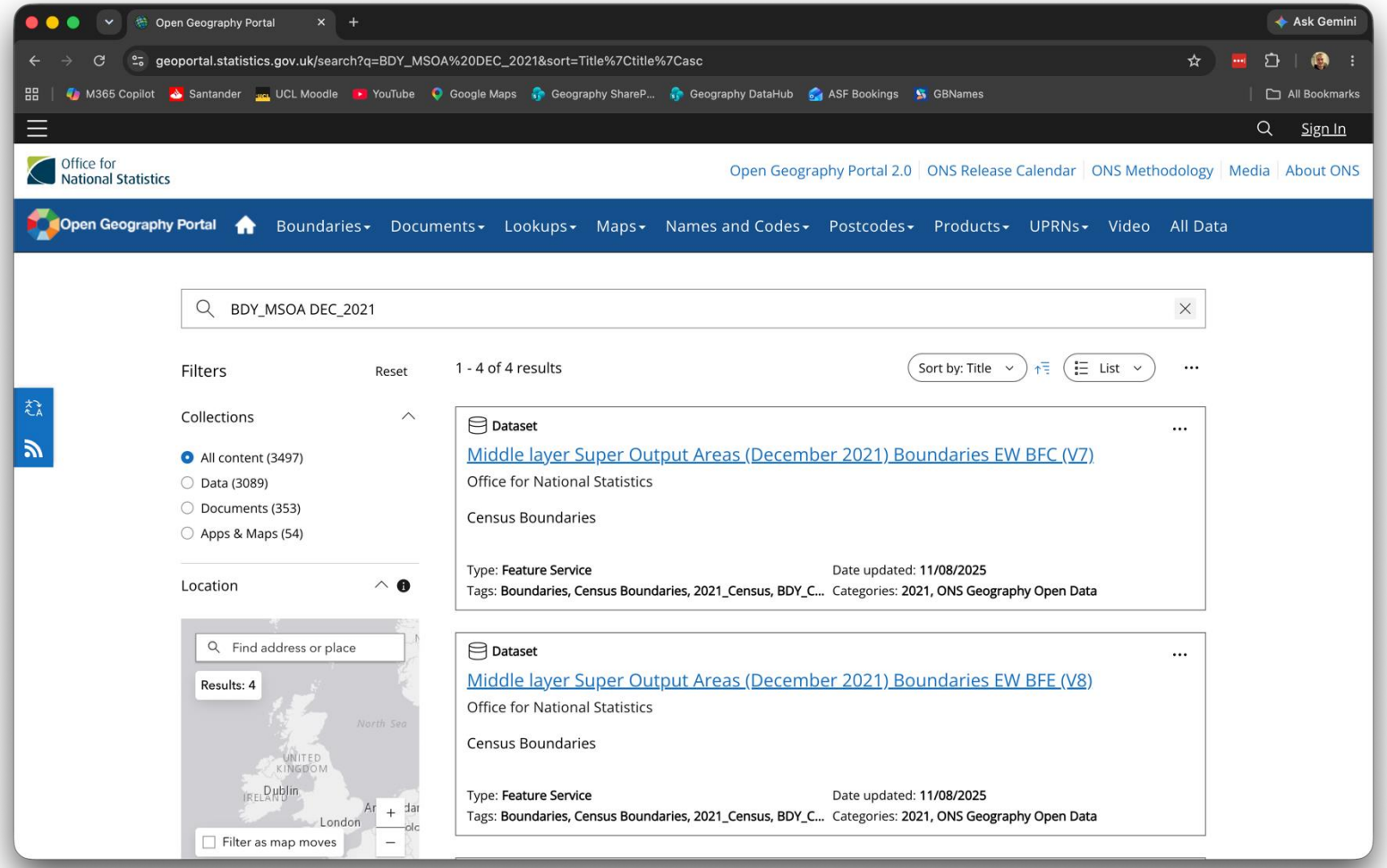
# nomis

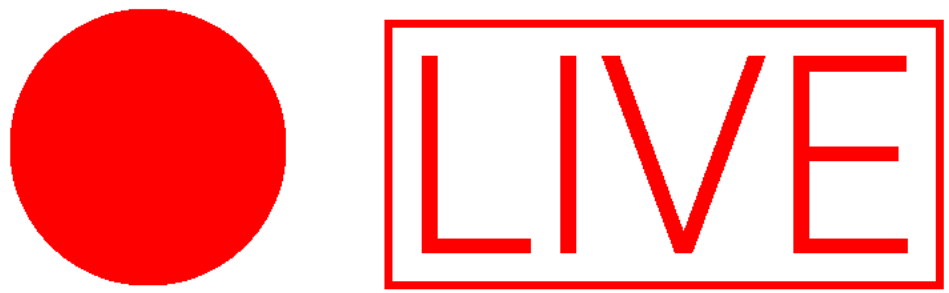

Nomis - Official Census and
Ask Gemini
nomisweb.co.uk
M365 Copilot
Santander
UCL Moodle
YouTube
Google Maps
Geography ShareP...
Geography DataHub
ASF Bookings
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Regional and National Profile
2024 Parliamentary Constituencies Profile
2011 Ward Profile (England & Wales only)
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View a 2021 Census report for a local area such as a ward, village or town. Includes information on the characteristics of people and households in the area.
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Go to 2011 Census local area report
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Search by keyword and geography type.
2011 Data catalogue
Browse by table type and number, or view by release.
2011 Search by topic (table finder)
Search by keyword and geography type.
Earlier censuses

# Office for National Statistics



# Open Geography Portal



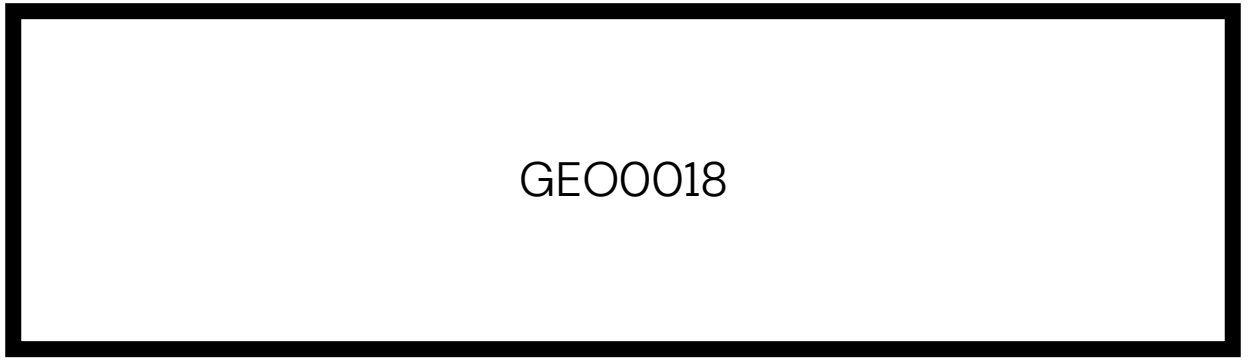

LIVE

# Attribute joins

# Attribute joins

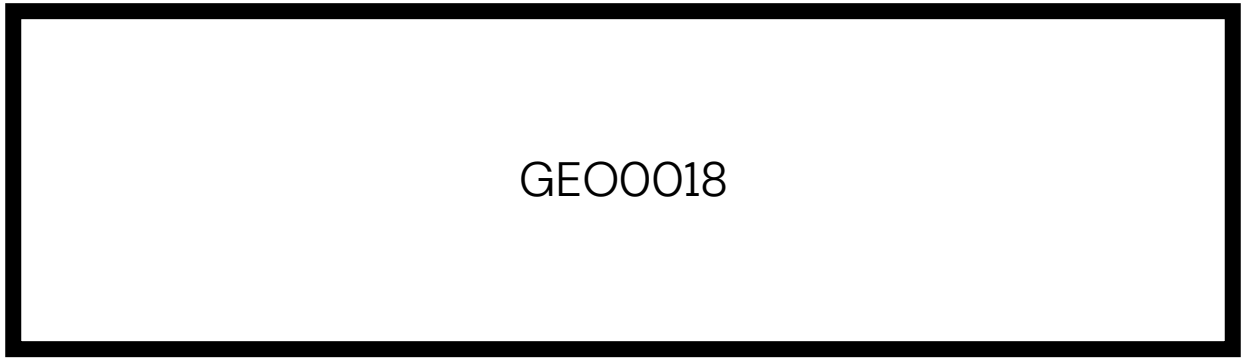
| GeoID | Population |
| --- | --- |
| GEO0018 | 540 |
| GEO0019 | 320 |

# Attribute joins




| FeatureID | GeoID |
| --- | --- |
| 0050 | GEO0018 |

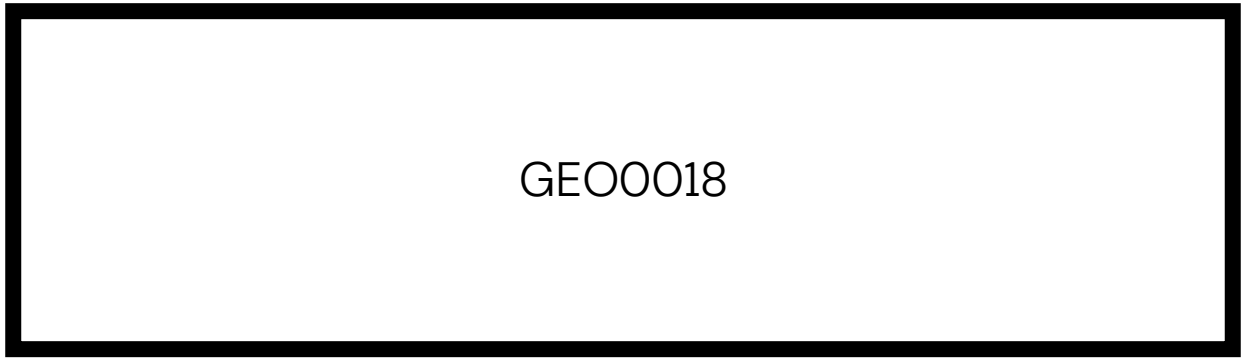
# Attribute joins




| FeatureID | GeoID |
|---|---|
| 0050 | GEO0018 |

| GeoID | Population |
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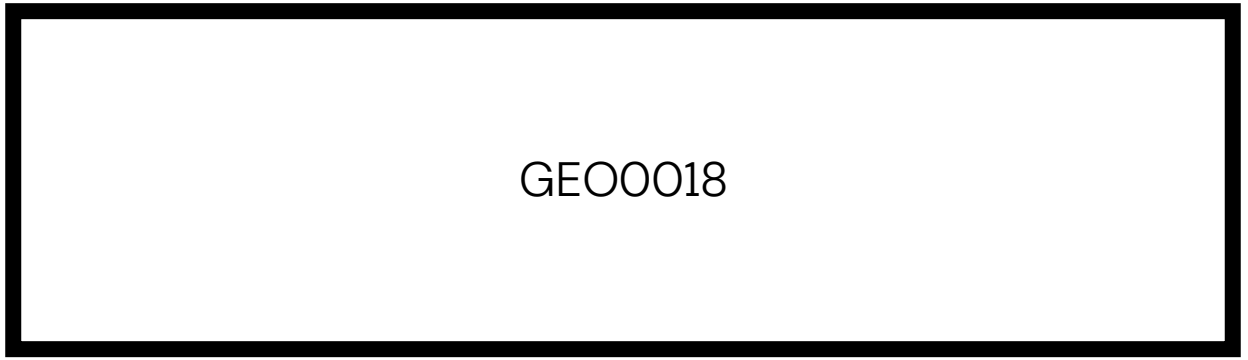
# Attribute joins




| FeatureID | GeoID |
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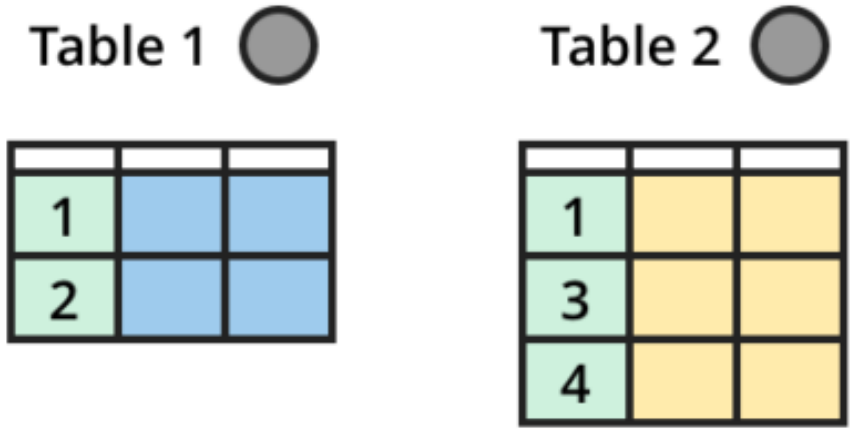
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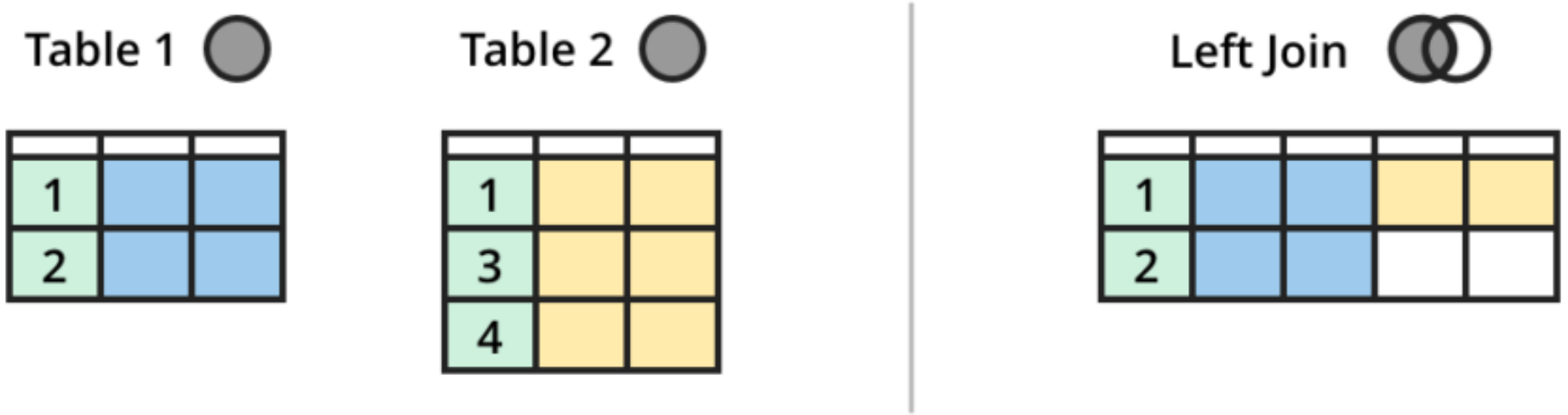

| FeatureID | GeoID | Population |
|---|---|---|
| 0050 | GEO0018 | 540 |

# Left joins




The Data School. 2019. *SQL Join Types Explained Visually*.

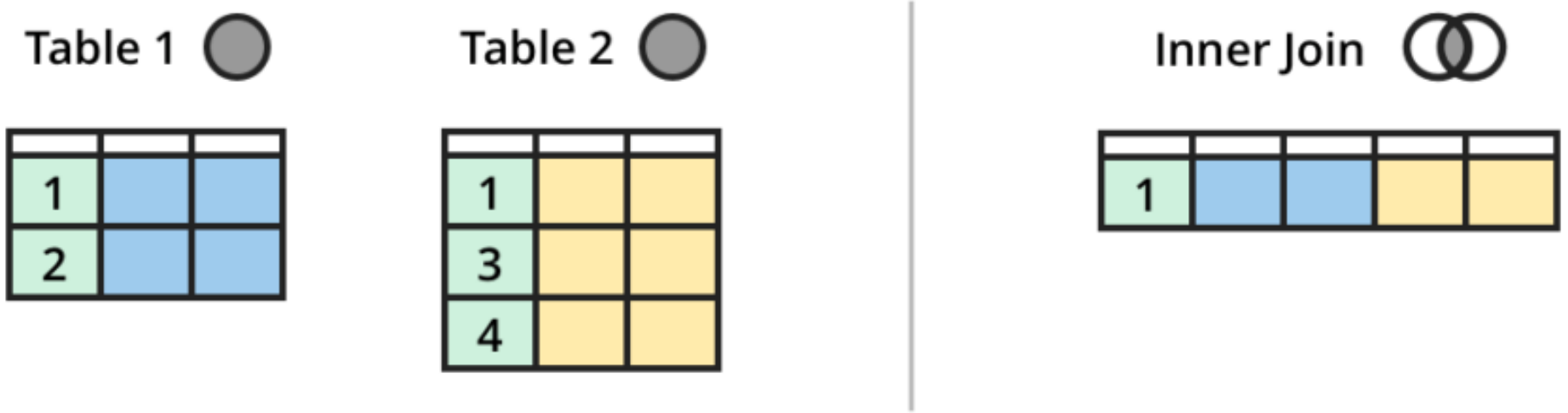
# Left joins




In QGIS every polygon stays; unmatched polygons get empty values.

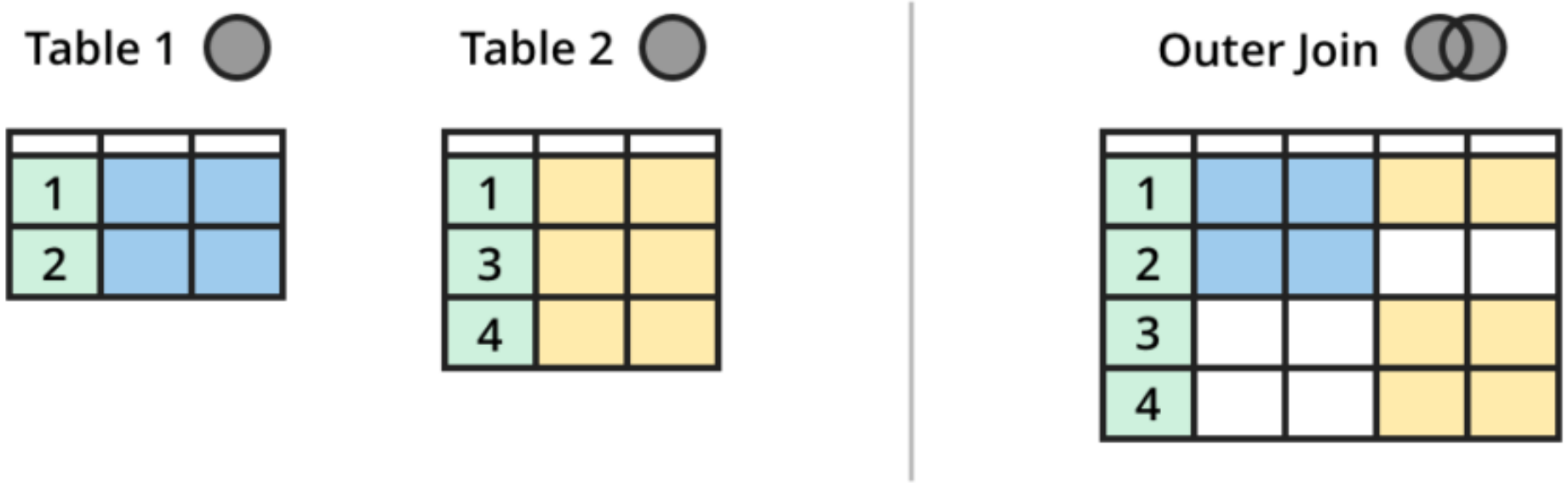
The Data School. 2019. *SQL Join Types Explained Visually*.

# Inner joins

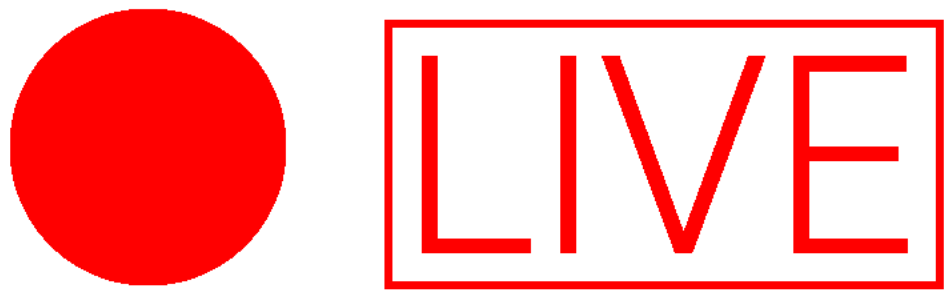



The Data School. 2019. *SQL Join Types Explained Visually*.

# Outer joins




The Data School. 2019. *SQL Join Types Explained Visually*.

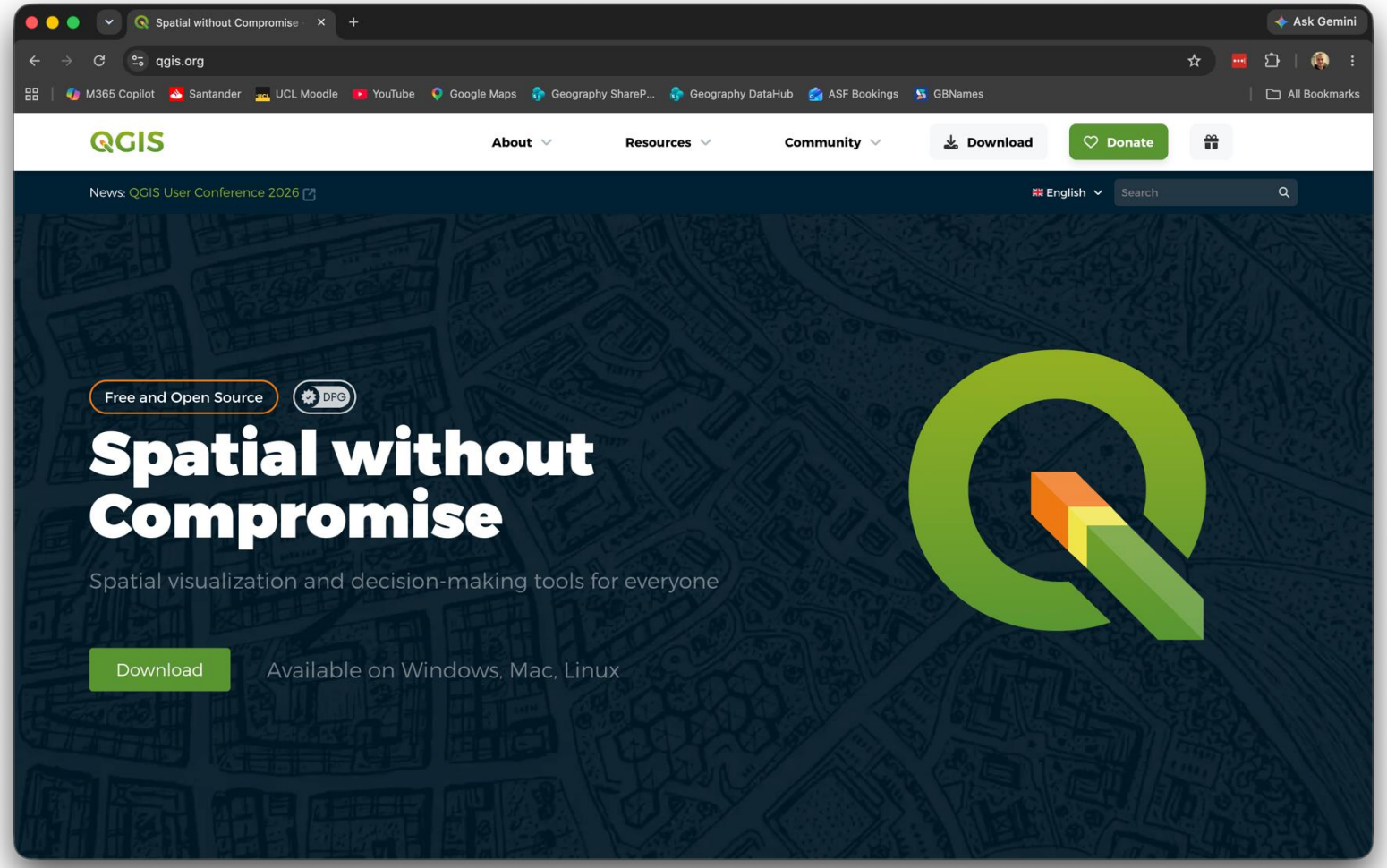

LIVE

# Summary

# Summary

- Many challenges and problems are inherently geographical.
- Spatial data visualisation is an essential part of (quantitative) human geography for exploring and communicating spatial patterns.
- A wide range of secondary data sources is available, covering topics such as demographics, environment, and socio-economic trends.
- Spatial data is represented as vector (points, lines, polygons) or raster (grid cells).
- An attribute join links a table to spatial features through a shared unique identifier.

# Summary


Spatial without Compromise
Ask Gemini
qgis.org
M365 Copilot
Santander
UCL Moodle
YouTube
Google Maps
Geography ShareP...
Geography DataHub
ASF Bookings
GBNames
All Bookmarks
QGIS
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Resources
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Free and Open Source
DPG
Spatial without Compromise
Spatial visualization and decision-making tools for everyone
Download
Available on Windows, Mac, Linux

# Assessment

To explore a geographic theme (e.g. gentrification, identity, migration) in the context of London, submitted as a 1,500-word report (50% of module grade):

1. Outline a research question and hypothesis (200-250 words).
2. Obtain data and identify appropriate methods to use (200-250 words).
3. Present and interpret results (700-800 words).
4. Briefly evaluate your approach (200-300 words).

Deadline 14th of January 2027 – 13h00.

# Questions

Justin van Dijk

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